Our experience in the study of ordinary series suggests that all these methods should produce the same sum if the numbers summed are all nonnegative, but that subtle differences are likely to emerge if we are required to add numbers both positive and negative.

In the exercises there are a number of problems that can be pursued to give a flavor for this kind of theory. At this stage in your studies it is important to grasp the fact that such questions arise. Later, when you have found a need to use these kinds of sums, you can develop the needed theory. The tools for developing that theory are just those that we have studied so far in this chapter.

Exercises

3.10.1 Decide on a meaning for the notion of a double series

$$\sum_{j,k=1}^{\infty} a_{jk} \tag{8}$$

and prove that if all the numbers a_{jk} are nonnegative then this converges if and only if

$$\sum_{(j,k)\in\mathbb{N}\times\mathbb{N}} a_{jk} \tag{9}$$

converges and that the values assigned to (8) and (9) are the same.

3.10.2 Decide on a meaning for the notion of an absolutely convergent double series

$$\sum_{j,k=1}^{\infty} a_{jk}$$

and prove that such a series is absolutely convergent if and only if

$$\sum_{(j,k)\in\mathbb{N}\times\mathbb{N}} a_{jk}$$

converges.

3.10.3 Show that the methods given in the text for forming a sum of a double sequence $\{a_{jk}\}$ are equivalent if all the numbers are nonnegative.

3.10.4 Show that the methods given in the text for forming a sum of a double sequence $\{a_{jk}\}$ are not equivalent in general.

3.10.5 What can you assert about the convergence or divergence of the double series

$$\sum_{j,k=1}^{\infty} \frac{1}{jk^4}?$$

3.10.6 What is the sum of the double series

$$\sum_{j,k=0}^{\infty} \frac{x^j y^k}{j!\,k!}?$$

3.11 Infinite Products

✂ *Enrichment section. May be omitted.*

In this chapter we studied, quite extensively, infinite sums. There is a similar theory for infinite products, a theory that has much in common with the theory of infinite sums. In this section we shall briefly give an account of this theory, partly to give a contrast and partly to introduce this important topic.

Similar to the notion of an infinite sum

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + a_4 + \dots$$

is the notion of an infinite product

$$\prod_{n=1}^{\infty} p_n = p_1 \times p_2 \times p_3 \times p_4 \times \dots$$

with a nearly identical definition. Corresponding to the concept of "partial sums" for the former will be the notion of "partial products" for the latter.

The main application of infinite series is that of series representations of functions. The main application of infinite products is exactly the same. Thus, for example, in more advanced material we will find a representation of the sin function as an infinite series

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

and also as an infinite product

$$\sin x = \left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{4\pi^2}\right)\left(1 - \frac{x^2}{9\pi^2}\right)\left(1 - \frac{x^2}{16\pi^2}\right)\dots.$$

The most obvious starting point for our theory would be to define an infinite product as the limit of the sequence of partial products in exactly the same way that an infinite sum is defined as the limit of the sequence of partial sums. But products behave differently from sums in one important regard: The number zero plays a peculiar role. This is why the definition we now give is slightly different than a first guess might suggest. Our goal is to define an infinite product in such a way that a product can be zero only if one of the factors is zero (just like the situation for finite products).

Definition 3.61: Let $\{b_k\}$ be a sequence of real numbers. We say that the infinite product

$$\prod_{k=1}^{\infty} b_k$$

converges if there is an integer N so that all $b_k \neq 0$ for $k > N$ and if

$$\lim_{M\to\infty} \prod_{k=N+1}^{M} b_k$$

exists *and is not zero*. For the value of the infinite product we take

$$\prod_{k=1}^{\infty} b_k = b_1 \times b_2 \times \dots b_N \times \lim_{M\to\infty} \prod_{k=N+1}^{M} b_k.$$

This definition guarantees us that a product of factors can be zero if and only if one of the factors is zero. This is the case for finite products, and we are reluctant to lose this.

Theorem 3.62: *A convergent product*

$$\prod_{k=1}^{\infty} b_k = 0$$

if and only if one of the factors is zero.

Proof. This is built into the definition and is one of its features. ■

We expect the theory of infinite products to evolve much like the theory of infinite series. We recall that a series $\sum_{k=1}^{n} a_k$ could converge only if $a_k \to 0$. Naturally, the product analog requires the terms to tend to 1.

Theorem 3.63: *A product*

$$\prod_{k=1}^{\infty} b_k$$

that converges necessarily has $b_k \to 1$ as $k \to \infty$.

Proof. This again is a feature of the definition, which would not be possible if we had not handled the zeros in this way. Choose N so that none of the factors b_k is zero for $k > N$. Then

$$b_n = \lim_{n\to\infty} \frac{\prod_{k=N+1}^{n} b_k}{\prod_{k=N+1}^{n-1} b_k} = 1$$

as required. ■

As a result of this theorem it is conventional to write all infinite products in the special form

$$\prod_{k=1}^{\infty} (1 + a_k)$$

and remember that the terms $a_k \to 0$ as $k \to \infty$ in a convergent product. Also, our assumption about the zeros allows for $a_k = -1$ only for finitely many values of k. The expressions $(1 + a_k)$ are called the "factors" of the product and the a_k themselves are called the "terms."

A close linkage with series arises because the two objects

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \prod_{k=1}^{\infty} (1 + a_k),$$

the series and the product, have much the same kind of behavior.

Theorem 3.64: *A product*

$$\prod_{k=1}^{\infty} (1 + a_k)$$

where all the terms a_k are positive is convergent if and only if the series $\sum_{k=1}^{\infty} a_k$ converges.

Proof. Here we use our usual criterion that has served us through most of this chapter: A sequence that is monotonic is convergent if and only if it is bounded.

Note that

$$a_1 + a_2 + a_3 + \cdots + a_n \leq (1+a_1)(1+a_2)(1+a_3) \times \cdots \times (1+a_n)$$

so that the convergence of the product gives an upper bound for the partial sums of the series. It follows that if the product converges so must the series.

In the other direction we have

$$(1+a_1)(1+a_2)(1+a_3) \times \cdots \times (1+a_n) \leq e^{a_1+a_2+a_3+\cdots+a_n}$$

and so the convergence of the series gives an upper bound for the partial products of the infinite product. It follows that if the series converges, so must the product. ■

Exercises

3.11.1 Give an example of a sequence of positive numbers $\{b_k\}$ so that

$$\lim_{n\to\infty} b_1 b_2 b_3 \ldots b_n$$

exists, but so that the infinite product

$$\prod_{n=1}^{\infty} b_k$$

nonetheless diverges.

3.11.2 Compute

$$\prod_{k=1}^{\infty} \left(1 - \frac{1}{k^2}\right).$$

3.11.3 In Theorem 3.64 we gave no relation between the value of the product

$$\prod_{k=1}^{\infty} (1+a_k)$$

and the value of the series $\sum_{k=1}^{\infty} a_k$ where all the terms a_k are positive. What is the best you can state?

3.11.4 For what values of p does the product

$$\prod_{n=1}^{\infty} \left(1 + \frac{1}{k^p}\right)$$

converge?

3.11.5 Show that

$$\prod_{k=1}^{\infty} (1+x^{2^k}) = (1+x^2) \times (1+x^4) \times (1+x^8) \times (1+x^{16}) \times \ldots$$

converges to $1/(1-x^2)$ for all $-1 < x < 1$ and diverges otherwise.

3.11.6 Find a Cauchy criterion for the convergence of infinite products.

3.11.7 A product

$$\prod_{k=1}^{\infty} (1+a_k)$$

is said to *converge absolutely* if the related product

$$\prod_{k=1}^{\infty} (1+|a_k|)$$

converges.

(a) Show that an absolutely convergent product is convergent.

(b) Show that an infinite product

$$\prod_{k=1}^{\infty}(1+a_k)$$

converges absolutely if and only if the series of its terms $\sum_{k=1}^{\infty} a_k$ converges absolutely.

(c) For what values of x does the product

$$\prod_{k=1}^{\infty}\left(1+\frac{x}{k}\right)$$

converge absolutely?

(d) For what values of x does the product

$$\prod_{k=1}^{\infty}\left(1+\frac{x}{k^2}\right)$$

converge absolutely?

(e) For what values of x does the product

$$\prod_{k=1}^{\infty}\left(1+x^k\right)$$

converge absolutely?

(f) Show that

$$\prod_{k=1}^{\infty}\left(1+\frac{(-1)^k}{k}\right)$$

converges but not absolutely.

3.11.8 Develop a theory that allows for the order of the factors in a product to be rearranged.

3.12 Challenging Problems for Chapter 3

3.12.1 If a_n is a sequence of positive numbers such that $\sum_{n=1}^{\infty} a_n$ diverges what (if anything) can you say about the following three series?

(a) $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$

(b) $\sum_{n=1}^{\infty} \frac{a_n}{1+na_n}$

(c) $\sum_{n=1}^{\infty} \frac{a_n}{1+n^2a_n}$

3.12.2 Prove the following variant on the Dirichlet Test 3.44: If $\{b_n\}$ is a sequence of bounded variation (cf. Exercise 3.5.12) that converges to zero and the partial sums of the series $\sum_{k=1}^{\infty} a_k$ are bounded, then the series $\sum_{k=1}^{\infty} a_k b_k$ converges.

3.12.3 Prove this variant on the Cauchy condensation test: If the terms of a series $\sum_{k=1}^{\infty} a_k$ are nonnegative and decrease monotonically to zero, then that series converges if and only if the series

$$\sum_{j=1}^{\infty}(2j+1)a_{j^2}$$

converges.

3.12.4 Prove this more general version of the Cauchy condensation test: If the terms of a series $\sum_{k=1}^{\infty} a_k$ are nonnegative and decrease monotonically to zero, then that series converges if and only if the related series

$$\sum_{j=1}^{\infty} (m_{j+1} - m_j) a_{m_j}$$

converges. Here $m_1 < m_2 < m_3 < m_4 < \dots$ is assumed to be an increasing sequence of integers and

$$m_{j+1} - m_j \le C\,(m_j - m_{j-1})$$

for some positive constant and all j.

3.12.5 For any two series of positive terms write

$$\sum_{k=1}^{\infty} a_k \preceq \sum_{k=1}^{\infty} b_k$$

if $a_k/b_k \to 0$ as $k \to \infty$.

(a) If both series converge, explain why this might be interpreted by saying that $\sum_{k=1}^{\infty} a_k$ is converging faster than $\sum_{k=1}^{\infty} b_k$.

(b) If both series diverge, explain why this might be interpreted by saying that $\sum_{k=1}^{\infty} a_k$ is diverging more slowly than $\sum_{k=1}^{\infty} b_k$.

(c) For convergent series is there any connection between

$$\sum_{k=1}^{\infty} a_k \preceq \sum_{k=1}^{\infty} b_k$$

and

$$\sum_{k=1}^{\infty} a_k \le \sum_{k=1}^{\infty} b_k ?$$

(d) For what values of p, q is

$$\sum_{k=1}^{\infty} \frac{1}{k^p} \preceq \sum_{k=1}^{\infty} \frac{1}{k^q} ?$$

(e) For what values of r, s is

$$\sum_{k=1}^{\infty} r^k \preceq \sum_{k=1}^{\infty} s^k ?$$

(f) Arrange the divergent series

$$\sum_{k=2}^{\infty} \frac{1}{k}\,,\ \sum_{k=2}^{\infty} \frac{1}{k\log k}\,,\ \sum_{k=2}^{\infty} \frac{1}{k\log(\log k)}\,,\ \sum_{k=2}^{\infty} \frac{1}{k\log(\log(\log k))} \cdots$$

into the correct order.

(g) Arrange the convergent series

$$\sum_{k=2}^{\infty} \frac{1}{k^p}\,,\ \sum_{k=2}^{\infty} \frac{1}{k(\log k)^p}\,,\ \sum_{k=2}^{\infty} \frac{1}{k\log k(\log(\log k))^p}\,,$$

$$\sum_{k=2}^{\infty} \frac{1}{k\log k(\log(\log k))(\log(\log(\log k)))^p} \cdots$$

into the correct order. Here $p > 1$.

(h) Suppose that $\sum_{k=1}^{\infty} b_k$ is a divergent series of positive numbers. Show that there is a series

$$\sum_{k=1}^{\infty} a_k \preceq \sum_{k=1}^{\infty} b_k$$

that also diverges (but more slowly).

(i) Suppose that $\sum_{k=1}^{\infty} a_k$ is a convergent series of positive numbers. Show that there is a series

$$\sum_{k=1}^{\infty} a_k \preceq \sum_{k=1}^{\infty} b_k$$

that also converges (but more slowly).

(j) How would you answer this question? Is there a "mother" of all divergent series diverging so slowly that all other divergent series can be proved to be divergent by a comparison test with that series?

SEE NOTE 62

3.12.6 This collection of exercises develops some convergence properties of *trigonometric series*; that is, series of the form

$$a_0/2 + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx). \tag{10}$$

(a) For what values of x does $\sum_{k=1}^{\infty} \frac{\sin kx}{k^2}$ converge?

(b) For what values of x does $\sum_{k=1}^{\infty} \frac{\sin kx}{k}$ converge?

(c) Show that the condition $\sum_{k=1}^{\infty} (|a_k| + |b_k|) < \infty$ ensures the absolute convergence of the trigonometric series (10) for all values of x.

SEE NOTE 63

3.12.7 Let $\{a_k\}$ be a decreasing sequence of positive real numbers with limit 0 such that

$$b_k = a_k - 2a_{k+1} + a_{k+2} \geq 0.$$

Prove that $\sum_{k=1}^{\infty} k b_k = a_1$.

SEE NOTE 64

3.12.8 Let $\{a_k\}$ be a monotonic sequence of real numbers such that $\sum_{k=1}^{\infty} a_k$ converges. Show that

$$\sum_{k=1}^{\infty} k(a_k - a_{k+1})$$

converges.

SEE NOTE 65

3.12.9 Show that every positive rational number can be obtained as the sum of a finite number of distinct terms of the harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \ldots.$$

SEE NOTE 66

3.12.10 Let $\sum_{k=1}^{\infty} x_k$ be a convergent series of positive numbers that is monotonically nonincreasing; that is, $x_1 \geq x_2 \geq x_3 \geq \ldots$. Let P denote the set of all real numbers that are sums of finitely or infinitely many terms of the series. Show that P is an interval if and only if

$$x_n \leq \sum_{k=n+1}^{\infty} x_k$$

for every integer n.

SEE NOTE 67

3.12.11 Let p_1, p_2, p_3, be a sequence of distinct points that is dense in the interval $(0,1)$. The points p_1, p_2, p_3, $\ldots$, p_{n-1} decompose the interval $[0,1]$ into n closed subintervals. The point p_n is an interior point of one of those intervals and decomposes that interval into two closed subintervals. Let a_n and b_n be the lengths of those two intervals. Prove that

$$\sum_{k=1}^{\infty} a_k b_k (a_k + b_k) = 3.$$

SEE NOTE 68

3.12.12 Let $\{a_n\}$ be a sequence of positive number such that the series $\sum_{k=1}^{\infty} a_k$ converges. Show that

$$\sum_{k=1}^{\infty} (a_k)^{n/(n+1)}$$

also converges.

SEE NOTE 69

3.12.13 Let $\{a_k\}$ be a sequence of positive numbers and suppose that

$$a_k \leq a_{2k} + a_{2k+1}$$

for all $k = 1,2,3,4,\ldots$. Show that $\sum_{k=1}^{\infty} a_k$ diverges.

SEE NOTE 70

3.12.14 If $\{a_k\}$ is a sequence of positive numbers for which $\sum_{k=1}^{\infty} a_k$ diverges, determine all values of p for which

$$\sum_{k=1}^{\infty} \frac{a_k}{(a_1 + a_2 + \cdots + a_k)^p}$$

converges.

SEE NOTE 71

3.12.15 Let $\{a_n\}$ be a sequence of real numbers converging to zero. Show that there must exist a monotonic sequence $\{b_n\}$ such that the series $\sum_{k=1}^{\infty} b_k$ diverges and the series $\sum_{k=1}^{\infty} a_k b_k$ is absolutely convergent.

SEE NOTE 72

Notes

[36]Exercise 3.2.2. Define $\sum_{i \in I} a_i$ for I with zero or one elements. Suppose it is defined for I with n elements. Define it for I with $n+1$ elements and show well defined.

[37]Exercise 3.2.4. The answer is yes if I and J are disjoint. Otherwise the correct formula would be

$$\sum_{i\in I\cup J} a_i + \sum_{i\in I\cap J} a_i = \sum_{i\in I} a_i + \sum_{i\in J} a_i.$$

[38]Exercise 3.2.8. Try to interpret the "difference" $\Delta s_k = s_{k+1} - s_k = a_{k+1}$ as the analog of a derivative.

[39]Exercise 3.2.11. Use a telescoping sum method. Even if you cannot remember your trigonometric identities you can work backward to see which one is needed. Check the formula for values of θ with $\sin\theta/2 = 0$ and see that it can be interpreted by taking limits.

[40]Exercise 3.3.1. This is similar to the statement that convergent sequences have unique limits. Try to imitate that proof.

[41]Exercise 3.3.2. This is similar to the statement that convergent sequences are bounded. Try to imitate that proof.

[42]Exercise 3.3.3. This is similar to the statement that monotone, bounded sequences are convergent. Try to imitate that proof.

[43]Exercise 3.3.9. Compare with the sum

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 2$$

given in the introduction to this chapter.

[44]Exercise 3.3.11. Here we are using, as elsewhere,

$$[X]^+ = \max\{X, 0\}$$

and

$$[X]^- = \max\{-X, 0\}$$

and note that

$$X = [X]^+ - [X]^- \text{ and } |X| = [X]^+ + [X]^-.$$

[45]Exercise 3.3.12. Note that the index set is

$$I = \mathbb{N} \times \mathbb{N}.$$

Thus we can study unordered sums of double sequences $\{a_{ij}\}$ in the form

$$\sum_{(i,j)\in\mathbb{N}\times\mathbb{N}} a_{ij}.$$

[46]Exercise 3.4.10. Handle the case where each $a_k \geq 0$ separately from the general case.

[47]Exercise 3.4.15. Using properties of the log function, you can view this series as a telescoping one.

[48]Exercise 3.4.16. Consider that

$$\frac{1}{r-1} - \frac{1}{r+1} = \frac{2}{r^2-1}.$$

[49]Exercise 3.4.24. Establish the inequalities

$$\sum_{k=1}^{2^n-1} \frac{1}{k^p} \leq \sum_{k=1}^{\infty} \frac{2^{k-1}}{\left(2^{k-1}\right)^p}$$

$$= \sum_{j=0}^{\infty} (2^{1-p})^j = \frac{2^{p-1}}{2^{p-1}-1}.$$

Conclude that the partial sums of the p-harmonic series for $p > 1$ are increasing and bounded. Explain now why the series must converge.

[50]Exercise 3.4.26. As a first step show that

$$\int_{2k\pi+\pi/4}^{2k\pi+3\pi/4} \frac{|\sin x|}{x}\,dx$$

$$\geq \frac{1}{\sqrt{2}} \int_{2k\pi+\pi/4}^{2k\pi+3\pi/4} \frac{1}{x}\,dx.$$

(Remember that in calculus an integral $\int_0^\infty$ is interpreted as $\lim_{X\to\infty} \int_0^X$.)

[51]Exercise 3.4.28. Establish that

$$\left| x - \sum_{i=1}^{n} \frac{k_i}{p^i} \right| \leq \frac{1}{p^n}.$$

[52]Exercise 3.5.5. Add up the terms containing p digits in the denominator. Note that our deletions leave only $8 \times 9^{p-1}$ of them. The total sum is bounded by

$$8(1/1 + 9/10 + 9^2/100 + \ldots) = 80.$$

[53]Exercise 3.5.8. Instead consider the series

$$\sum_{k=1}^{\infty} [a_k]^+ \text{ and } \sum_{k=1}^{\infty} [a_k]^-$$

where

$$[X]^+ = \max\{X, 0\}$$

and

$$[X]^- = \max\{-X, 0\}$$

and note that

$$X = [X]^+ - [X]^- \text{ and } |X| = [X]^+ + [X]^-.$$

[54]Exercise 3.5.15. Use the Cauchy-Schwarz inequality.

[55]Exercise 3.5.16. Use the Cauchy-Schwarz inequality.

[56]Exercise 3.6.3. The answer for (d) is $x < 1/e$.

[57]Exercise 3.6.5. Only one condition is sufficient to supply divergence. Give a proof for that one and counterexamples for the three others. Here is an idea that may help: Let $a_k = 0$ for all values of k except if $k = 2^m$ for some m in which case $a_k = 1/\sqrt{k}$. Note that $\limsup_{k\to\infty} \sqrt{k} a_k = 1$ in this case and that $\sum_{k=1}^{\infty} a_k$ will converge.

[58]Exercise 3.6.22. The exact value of γ, called *Euler's constant*, is not needed in the problem; it is approximately .5772156.

[59]Exercise 3.6.24. The integral test should occur to you while thinking of this problem. Start by checking that

$$\sum_{k=1}^{\infty} F'(k)$$

converges if and only if

$$\lim_{X\to\infty} F(X)$$

exists. Find similar statements for the other series.

[60]Exercise 3.7.13. Imitate the proof of the first part of Theorem 3.49 but arrange for the partial sums to go larger than α before inserting a term q_k. You must take the *first* opportunity to insert q_k when this occurs.

[61]Exercise 3.9.6. The name "Tauberian theorem" was coined by Hardy and Littlewood after a result of Alfred Tauber (1866–1942?). The date of his death is unknown; all that is certain is that he was sent by the Nazis to Theresienstadt concentration camp on June 28, 1942.

[62]Exercise 3.12.5. For (h) consider the series $\sum_{k=1}^{\infty}(s_{k+1}-s_k)/s_{k+1}$ where s_k is the sequence of partial sums of the series given.

[63]Exercise 3.12.6. For (b) use Abel's method and the computation in Exercise 3.2.11. Further treatment of some aspects of trigonometric series may be found in Section 10.8.

[64]Exercise 3.12.7. This is from the 1948 Putnam Mathematical Competition.

[65]Exercise 3.12.8. This is from the 1952 Putnam Mathematical Competition.

[66]Exercise 3.12.9. This is from the 1954 Putnam Mathematical Competition.

[67]Exercise 3.12.10. This is from the 1955 Putnam Mathematical Competition.

[68]Exercise 3.12.11. This is from the 1964 Putnam Mathematical Competition.

[69]Exercise 3.12.12. This is from the 1988 Putnam Mathematical Competition.

[70]Exercise 3.12.13. This is from the 1994 Putnam Mathematical Competition.

[71]Exercise 3.12.14. Problem posed by A. Torchinsky in *Amer. Math. Monthly*, **82** (1975), p. 936.

[72]Exercise 3.12.15. Problem posed by Jan Mycielski in *Amer. Math. Monthly*, **83** (1976), p. 284.

Chapter 4

SETS OF REAL NUMBERS

4.1 Introduction

Modern set theory and the world it has opened to mathematics has its origins in a problem in analysis. A young Georg Cantor in 1870 began to attack a problem given to him by his senior colleague Edward Heine, who worked at the same university. (We shall see Heine playing a key role in some ideas of this chapter too.)

The problem was to determine if the equation

$$\tfrac{1}{2}a_0 + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx) = 0 \tag{1}$$

must imply that all the coefficients of the series, the $\{a_k\}$ and the $\{b_k\}$ are zero. Cantor solved this using the methods of his time. It was a good achievement, but not the one that was to make him famous. What he did next was to ask, as any good mathematician would, whether his result could be generalized. Suppose that the series (1) converges to zero for all x except possibly for those in a given set E. If this set E is very small, then perhaps, the coefficients of the series should also have to be all zero.

The nature of these exceptional sets (nowadays called sets of uniqueness) required a language and techniques that were entirely new. Previously a number of authors had needed a language to describe sets that arose in various problems. What was used at the time was limited, and few interesting examples of sets were available. Cantor went beyond these, introducing a new collection of ideas that are now indispensable to analysis. We shall encounter in this chapter many of the notions that arose then: accumulation points, derived sets, countable sets, dense sets, nowhere dense sets.

Incidentally, Cantor never did finish his problem of describing the sets of uniqueness, as the development of the new set theory was more important and consumed his energies. In fact, the problem remains unsolved, although much interesting information about the nature of sets of uniqueness has been discovered.

The theory of sets that Cantor initiated has proved to be fundamental to all of mathematics. Very quickly the most talented analysts of that time began applying his ideas to the theory of functions, and by now this material is essential to an understanding of the subject. This chapter contains the most basic material. In Chapter 6 we will need some further concepts.

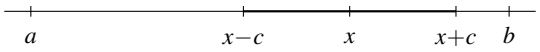


Figure 4.1. Every point in (a,b) is an interior point.

4.2 Points

In our studies of analysis we shall often need to have a language that describes sets of points and the points that belong to them. That language did not develop until late in the nineteenth century, which is why the early mathematicians had difficulty understanding some problems.

For example, consider the set of solutions to an equation

$$f(x) = 0$$

where f is some well-behaved function. In the simplest cases (e.g., if f is a polynomial function) the solution set could be empty or a finite number of points. There is no difficulty there. But in more general settings the solution set could be very complicated indeed. It may have points that are "isolated," points appearing in clusters, or it may contain intervals or merely fragments of intervals. You can see that we even lack the words to describe the possibilities.

The ideas in this section are all very geometric. Try to draw mental images that depict all of these ideas to get a feel for the definitions. The definitions themselves should be remembered but may prove hard to remember without some associated picture.

The simplest types of sets are intervals. We call

$$[a,b] = \{x : a \leq x \leq b\}$$

a closed interval, and

$$(a,b) = \{x : a < x < b\}$$

an open interval. The other sets that we often consider are the sets $\mathbb{N}$ of natural numbers, $\mathbb{Q}$ of rational numbers, and $\mathbb{R}$ of all real numbers. Use these in your pictures, as well as sets obtained by combining them in many ways.

4.2.1 Interior Points

Every point inside an open interval $I = (a,b)$ has the feature that there is a smaller open interval centered at that point that is also inside I. Thus if $x \in (a,b)$ then for any positive number c that is small enough

$$(x-c, x+c) \subset (a,b).$$

Indeed the arithmetic to show this is easy (and a picture makes it transparent). Let c be any positive number that is smaller than the shortest distance from x to either a or b. Then $(x-c, x+c) \subset (a,b)$. (See Figure 4.1.)

Note. Often we use the following suggestive language. An open interval that contains a point x is said to be a *neighborhood* of x. Thus each point in (a,b) possesses a neighborhood, indeed many neighborhoods, that lie entirely inside the set I. On occasion the point x itself is excluded from the neighborhood: We say that an interval (c,d) is a neighborhood of x if x belongs to the interval and we say that the set $(c,d) \setminus \{x\}$ is a *deleted neighborhood*. This is just the interval with the point x removed.

We can distinguish between points that are merely in a set and points that are more deeply inside the set. The word chosen to convey this image of "inside" is *interior*.

Definition 4.1: (Interior Point) Let E be a set of real numbers. Any point x that belongs to E is said to be *an interior point of* E provided that some interval

$$(x-c,x+c) \subset E.$$

Thus an interior point of the set E is not merely *in the set* E; it is, so to speak, deep inside the set, at a positive distance at least c away from every point that does not belong to E.

Example 4.2: The following examples are immediate if a picture is sketched.

1. Every point x of an open interval (a,b) is an interior point.
2. Every point x of a closed interval $[a,b]$, except the two endpoints a and b, is an interior point.
3. The set of natural numbers $\mathbb{N}$ has no interior points whatsoever.
4. Every point of $\mathbb{R}$ is an interior point.
5. No point of the set of rational numbers $\mathbb{Q}$ is an interior point. [This is because any interval $(x-c,x+c)$ must contain both rational numbers and irrational numbers and, hence, can never be a subset of $\mathbb{Q}$.]

In each case, we should try to find the interval $(x-c,x+c)$ inside the set or explain why there can be no such interval. ◀

4.2.2 Isolated Points

Most sets that we consider will have infinitely many points. Certainly any interval (a,b) or $[a,b]$ has infinitely many points. The set $\mathbb{N}$ of natural numbers also has infinitely many points, but as we look closely at any one of these points we see that each point is all alone, at a certain distance away from every other point in the set. We call these points *isolated points* of the set.

Definition 4.3: (Isolated Point) Let E be a set of real numbers. Any point x that belongs to E is said to be *an isolated point of* E provided that for some interval $(x-c,x+c)$

$$(x-c,x+c) \cap E = \{x\}.$$

Thus an isolated point of the set E is in the set E but has no close neighbors who are also in E. It is at some positive distance at least c away from every other point that belongs to E.

Example 4.4: As before, the examples are immediate if a picture is sketched.

1. No point x of an open interval (a,b) is an isolated point.
2. No point x of a closed interval $[a,b]$ is an isolated point.
3. Every point belonging to the set of natural numbers $\mathbb{N}$ is an isolated point.
4. No point of $\mathbb{R}$ is isolated.
5. No point of $\mathbb{Q}$ is isolated.

In each case, we should try to find the interval $(x-c,x+c)$ that meets the set at no other point or show that there is none. ◀

4.2.3 Points of Accumulation

Most sets that we consider will have infinitely many points. While the isolated points are of interest on occasion, more than likely we would be interested in points that are not isolated. These points have the property that every containing interval contains many points of the set. Indeed we are interested in any point x with the property that the intervals $(x-c,x+c)$ meet the set E at infinitely many points. This could happen even if x itself does not belong to E. We call these points *accumulation points* of the set. An accumulation point need not itself belong to the set.

Definition 4.5: (Accumulation Point) Let E be a set of real numbers. Any point x (not necessarily in E) is said to be *an accumulation point of E* provided that for every $c > 0$ the intersection

$$(x-c,x+c) \cap E$$

contains infinitely many points.

Thus an accumulation point of E is a point that may or may not itself belong to E and that has many close neighbors who are in E.

Note. The definition requires that for all $c > 0$ the intersection

$$(x-c,x+c) \cap E$$

contains infinitely many points of E. In checking for an accumulation point it may be preferable merely to check that there is at least one point in this intersection (other than possibly x itself). If there is always at least one point, then there must in fact be infinitely many (Exercise 4.2.18).

Example 4.6: Yet again, the examples are immediate if a picture is sketched.

1. Every point of an open interval (a,b) is a an accumulation point of (a,b). Moreover, the two endpoints a and b are also accumulation points of (a,b) [although they do not belong themselves to (a,b))].

2. Every point of a closed interval $[a,b]$ is an accumulation point of (a,b). No point outside can be.

3. No point at all is an accumulation point of the set of natural numbers $\mathbb{N}$.

4. Every point of $\mathbb{R}$ is an accumulation point.

5. Every point on the real line, both rational and irrational, is an accumulation point of the set $\mathbb{Q}$.

◀

4.2.4 Boundary Points

The intervals (a,b) and $[a,b]$ have what appears to be an "edge". The points a and b mark the boundaries between the inside of the set (i.e., the interior points) and the "outside" of the set. This inside/outside language with an idea of a boundary between them is most useful but needs a precise definition.

Definition 4.7: (Boundary Point) Let E be a set of real numbers. Any point x (not necessarily in E) is said to be *a boundary point of E* provided that every interval $(x-c,x+c)$ contains at least one point of E and also at least one point that does not belong to E.

This definition is easy to apply to the intervals (a,b) and $[a,b]$ but harder to imagine for general sets. For these intervals the only points that are immediately seen to satisfy the definition are the two endpoints that we would have naturally said to be at the boundary.

Example 4.8: The examples are not all transparent but require careful thinking about the definition.

1. The two endpoints a and b are the only boundary points of an open interval (a,b).

2. The two endpoints a and b are the only boundary points of a closed interval $[a,b]$.

3. Every point in the set $\mathbb{N}$ of natural numbers is a boundary point.

4. No point at all is boundary point of the set $\mathbb{R}$.

5. Every point on the real line, both rational and irrational, is a boundary point of the set $\mathbb{Q}$. (Think for a while about this one!)

◀

Exercises

4.2.1 Determine the set of interior points, accumulation points, isolated points, and boundary points for each of the following sets:

(a) $\{1, 1/2, 1/3, 1/4, 1/5, \dots\}$

(b) $\{0\} \cup \{1, 1/2, 1/3, 1/4, 1/5, \dots\}$

(c) $(0,1) \cup (1,2) \cup (2,3) \cup (3,4) \cup \cdots \cup (n, n+1) \cup \dots$

(d) $(1/2,1) \cup (1/4,1/2) \cup (1/8,1/4) \cup (1/16,1/8) \cup \dots$

(e) $\{x : |x - \pi| < 1\}$

(f) $\{x : x^2 < 2\}$

(g) $\mathbb{R} \setminus \mathbb{N}$

(h) $\mathbb{R} \setminus \mathbb{Q}$

4.2.2 Give an example of each of the following or explain why you think such a set could not exist.

(a) A nonempty set with no accumulation points and no isolated points

(b) A nonempty set with no interior points and no isolated points

(c) A nonempty set with no boundary points and no isolated points

4.2.3 Show that every interior point of a set must also be an accumulation point of that set, but not conversely.

4.2.4 Show that no interior point of a set can be a boundary point, that it is possible for an accumulation point to be a boundary point, and that every isolated point must be a boundary point.

4.2.5 Let E be a nonempty set of real numbers that is bounded above but has no maximum. Let $x = \sup E$. Show that x is a point of accumulation of E. Is it possible for x to also be an interior point of E? Is x a boundary point of E?

4.2.6 State and solve the version of Exercise 4.2.5 that would use the infimum in place of the supremum.

4.2.7 Let A be a set and $B = \mathbb{R} \setminus A$. Show that every boundary point of A is also a boundary point of B.

4.2.8 Let A be a set and $B = \mathbb{R} \setminus A$. Show that every boundary point of A is a point of accumulation of A or else a point of accumulation of B, perhaps both.

4.2.9 Must every boundary point of a set be also an accumulation point of that set?

4.2.10 Show that every accumulation point of a set that does not itself belong to the set must be a boundary point of that set.

4.2.11 Show that a point x is not an interior point of a set E if and only if there is a sequence of points $\{x_n\}$ converging to x and no point $x_n \in E$.

4.2.12 Let A be a set and $B = \mathbb{R} \setminus A$. Show that every interior point of A is not an accumulation point of B.

4.2.13 Let A be a set and $B = \mathbb{R} \setminus A$. Show that every accumulation point of A is not an interior point of of B.

4.2.14 Give an example of a set that has the set $\mathbb{N}$ as its set of accumulation points.

4.2.15 Show that there is no set which has the interval $(0,1)$ as its set of accumulation points.

4.2.16 Show that there is no set which has the set $\mathbb{Q}$ as its set of accumulation points.

4.2.17 Give an example of a set that has the set

$$E = \{0\} \cup \{1, 1/2, 1/3, 1/4, 1/5, \dots\}$$

as its set of accumulation points.

4.2.18 Show that a point x is an accumulation point of a set E if and only if for every $\varepsilon > 0$ there are at least two points belonging to the set $E \cap (x-\varepsilon, x+\varepsilon)$.

4.2.19 Suppose that $\{x_n\}$ is a convergent sequence converging to a number L and that $x_n \neq L$ for all n. Show that the set

$$\{x : x = x_n \text{ for some } n\}$$

has exactly one point of accumulation, namely L. Of what importance was the assumption that $x_n \neq L$ for all n for this exercise?

4.2.20 Let E be a set and $\{x_n\}$ a sequence of distinct elements of E. Suppose that $\lim_{n\to\infty} x_n = x$. Show that x is a point of accumulation of E.

4.2.21 Let E be a set and $\{x_n\}$ a sequence of points, not necessarily elements of E. Suppose that $\lim_{n\to\infty} x_n = x$ and that x is an interior point of E. Show that there is an integer N so that $x_n \in E$ for all $n \geq N$.

4.2.22 Let E be a set and $\{x_n\}$ a sequence of elements of E. Suppose that

$$\lim_{n\to\infty} x_n = x$$

and that x is an isolated point of E. Show that there is an integer N so that $x_n = x$ for all $n \geq N$.

4.2.23 Let E be a set and $\{x_n\}$ a sequence of distinct points, not necessarily elements of E. Suppose that $\lim_{n\to\infty} x_n = x$ and that $x_{2n} \in E$ and $x_{2n+1} \notin E$ for all n. Show that x is a boundary point of E.

4.2.24 If E is a set of real numbers, then E', called the *derived set* of E, denotes the set of all points of accumulation of E. Give an example of each of the following or explain why you think such a set could not exist.

(a) A nonempty set E such that $E' = E$

(b) A nonempty set E such that $E' = \emptyset$

(c) A nonempty set E such that $E' \neq \emptyset$ but $E'' = \emptyset$

(d) A nonempty set E such that $E', E'' \neq \emptyset$ but $E''' = \emptyset$

(e) A nonempty set E such that $E', E'', E''', \dots$ are all different

(f) A nonempty set E such that $(E \cup E')' \neq (E \cup E')$

4.2.25 Show that there is no set with uncountably many isolated points.

SEE NOTE 73

4.3 Sets

We now begin a classification of sets of real numbers. Almost all of the concepts of analysis (limits, derivatives, integrals, etc.) can be better understood if a classification scheme for sets is in place. By far the most important notions are those of closed sets and open sets. This is the basis for much advanced mathematics and leads to the subject known as topology, which is fundamental to an understanding of many areas of mathematics. On the real line we can master open and closed sets and describe precisely what they are.

4.3.1 Closed Sets

In many parts of mathematics the word "closed" is used to indicate that some operation stays within a system. For example, the set of natural numbers $\mathbb{N}$ is closed under addition and multiplication (any sum or product of two of them is yet another) but not closed under subtraction or division (2 and 3 are natural numbers, but $2-3$ and $3/2$ are not). This same word was employed originally to indicate sets of real numbers that are "closed" under the operation of taking points of accumulation. If all points of accumulation turn out to be in the set, then the set is said to be closed. This terminology has survived and become, perhaps, the best known usage of the word "closed."

Definition 4.9: (Closed) Let E be a set of real numbers. The set E is said to be *closed* provided that every accumulation point of E belongs to the set E.

Thus a set E is not closed if there is some accumulation point of E that does not belong to E. In particular, a set with no accumulation points would have to be closed since there is no point that needs to be checked.

Example 4.10: The examples are immediate since we have previously described all of the accumulation points of these sets.

1. The empty set $\emptyset$ is closed since it contains all of its accumulation points (there are none).
2. The open interval (a,b) not closed because the two endpoints a and b are accumulation points of (a,b) and yet they do not belong to the set.
3. The closed interval $[a,b]$ is closed since only points that are already in the set are accumulation points.
4. The set of natural numbers $\mathbb{N}$ is closed because it has no points of accumulation.
5. The real line $\mathbb{R}$ is closed since it contains all of its accumulation points, namely every point.
6. The set of rational numbers $\mathbb{Q}$ is not closed. Every point on the real line, both rational and irrational, is an accumulation point of $\mathbb{Q}$, but the set fails to contain any irrationals.

◀

The Closure of a Set If a set is not closed it is because it neglects to contain points that "should" be there since they are accumulation points but not in the set. On occasions it is best to throw them in and consider a larger set composed of the original set together with the offending accumulation points that may not have belonged originally to the set.

Definition 4.11: (Closure) Let E be any set of real numbers and let E' denote the set of all accumulation points of E. Then the set

$$\overline{E} = E \cup E'$$

is called the *closure* of the set E.

For example, $\overline{(a,b)} = [a,b]$, $\overline{[a,b]} = [a,b]$, $\overline{\mathbb{N}} = \mathbb{N}$, and $\overline{\mathbb{Q}} = \mathbb{R}$. Each of these is an easy observation since we know what the points of accumulation of these sets are.

4.3.2 Open Sets

Originally, the word "open" was used to indicate a set that was not closed. In time it was realized that this is a waste of terminology, since the class of "not closed sets" is not of much general interest. Instead the word is now used to indicate a contrasting idea, an idea that is not quite an opposite—just at a different extreme. This may be a bit unfortunate since now a set that is not open need not be closed. Indeed some sets can be both open and closed, and some sets can be both not open and not closed.

Definition 4.12: (Open) Let E be a set of real numbers. Then E is said to be *open* if every point of E is also an interior point of E.

Thus every point of E is not merely a point *in the set* E; it is, so to speak, deep inside the set. For each point x_0 of E there is some positive number δ and all points outside E are at least a distance δ away from x_0. Note that this means that an open set cannot contain any of its boundary points.

Example 4.13: These examples are immediate since we have seen them before in the context of interior points in Section 4.2.1.

1. The empty set $\emptyset$ is open since it contains no points that are not interior points of the set. (This is the first example of a set that is both open and closed.)

2. The open interval (a,b) is open since every point x of an open interval (a,b) is an interior point.

3. The closed interval $[a,b]$ is not open since there are points in the set (namely the two endpoints a and b) that are in the set and yet are not interior points.

4. The set of natural numbers $\mathbb{N}$ has no interior points and so this set is not open; all of its points fail to be interior points.

5. Every point of $\mathbb{R}$ is an interior point and so $\mathbb{R}$ is open. (Remember, $\mathbb{R}$ is also closed so it is both open and closed. Note that $\mathbb{R}$ and $\emptyset$ are the only examples of sets that are both open and closed.)

6. No point of the set of rational numbers $\mathbb{Q}$ is an interior point and so $\mathbb{Q}$ definitely fails to be open.

◀

The Interior of a Set If a set is not open it is because it contains points that "shouldn't" be there since they are not interior. On occasions it is best to throw them away and consider a smaller set composed entirely of the interior points.

Definition 4.14: (Interior) Let E be any set of real numbers. Then the set

$$\text{int}(E)$$

denotes the set of all interior points of E and is called the *interior* of the set E.

For example, $\text{int}((a,b)) = (a,b)$, $\text{int}([a,b]) = (a,b)$, $\text{int}(\mathbb{N}) = \emptyset$, and $\text{int}(\mathbb{Q}) = \emptyset$. Each of these is an easy observation since we know what the interior points of these sets are.

Component Intervals of Open Sets Think of the most general open set G that you can. A first feeble suggestion might be any open interval $G = (a,b)$. We can do a little better. How about the union of two of these

$$G = (a,b) \cup (c,d)?$$

If these are disjoint, then we would tend to think of G as having two "components." It is easy to see that every point is an interior point. We need not stop at two component intervals; any number would work:

$$G = (a_1,b_1) \cup (a_2,b_2) \cup (a_3,b_3) \cup \cdots \cup (a_n,b_n).$$

The argument is the same and elementary. If x is a point in this set, then x is an interior point. Indeed we can form the union of a sequence of such open intervals and it is clear that we shall obtain an open set. For a specific example consider

$$(-\infty,-3) \cup (1/2,1) \cup (1/8,1/4) \cup (1/32,1/16) \cup (1/128,1/64) \cup \ldots.$$

At this point our imagination stalls and it is hard to come up with any more examples that are not obtained by stringing together open intervals in exactly this way. This suggests that, perhaps, all open sets have this structure. They are either open intervals or else a union of a sequence of open intervals. This theorem characterizes all open sets of real numbers and reveals their exact structure.

Theorem 4.15: *Let G be a nonempty open set of real numbers. Then there is a unique sequence (finite or infinite) of disjoint, open intervals*

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),\ldots,(a_n,b_n),\ldots$$

called the component intervals of *G such that*

$$G = (a_1,b_1) \cup (a_2,b_2) \cup (a_3,b_3) \cup \cdots \cup (a_n,b_n) \cup \ldots.$$

Proof. Take any point $x \in G$. We know that there must be some interval (a,b) containing the point x and contained in the set G. This is because G is open and so every point in G is an interior point. We need to take the largest such interval. The easiest way to describe this is to write

$$\alpha = \inf\{t : (t,x) \subset G\}$$

and

$$\beta = \sup\{t : (x,t) \subset G\}.$$

Note that $\alpha < x < \beta$. Then

$$I_x = (\alpha, \beta)$$

is called the *component* of G containing the point x. (It is possible here for $\alpha = -\infty$ or $\beta = \infty$.)

One feature of components that we require is this: If x and y belong to the same component, then

$$I_x = I_y$$

If x and y do not belong to the same component, then I_x and I_y have no points in common. This is easily checked (Exercise 4.3.21).

There remains the task of listing the components as the theorem requires. If the collection

$$\{I_x : x \in G\}$$

is finite, then this presents no difficulties. If it is infinite we need a clever strategy.

Let r_1, r_2, r_3, ... be a listing of all the rational numbers contained in the set G. We construct our list of components of G by writing for the first step

$$(a_1,b_1) = I_{r_1}.$$

The second component must be disjoint from this first component. We cannot simply choose I_{r_2} since if r_2 belongs to (a_1,b_1), then in fact

$$(a_1,b_1) = I_{r_1} = I_{r_2}.$$

Instead we travel along the sequence r_1, r_2, r_3, ... until we reach the first one, say r_{m_2}, that does not already belong to the interval (a_1,b_1). This then serves to define our next interval:

$$(a_2,b_2) = I_{r_{m_2}}.$$

If there is no such point, then the process stops. This process is continued inductively resulting in a sequence of open intervals:

$$(a_1,b_1),(a_2,b_2),(a_3,b_3),\ldots,(a_n,b_n),\ldots,$$

which may be infinite or finite. At the kth stage a point r_{m_k} is selected so that r_{m_k} does not belong to any component thus far selected. If this cannot be done, then the process stops and produces only a finite list of components.

The proof is completed by checking that (i) every point of G is in one of these intervals, (ii) every point in one of these intervals belongs to G, and (iii) the intervals in the sequence must be disjoint.

For (i) note that if $x \in G$, then there must be rational numbers in the component I_x. Indeed there is a first number r_k in the list that belongs to this component. But then $x \in I_{r_k}$ and so we must have chosen this interval I_{r_k} at some stage. Thus x does belong to one of these intervals.

For (ii) note that if x is in G, then $I_x \subset G$. Thus every point in one of the intervals belongs to G.

For (iii) consider some pair of intervals in the sequence we have constructed. The later one chosen was required to have a point r_{m_k} that did not belong to any of the preceding choices. But that means then that the new component chosen is disjoint from all the previous ones.

This completes the checking of the details and so the proof is done. ■

Exercises

4.3.1 Is it true that a set, all of whose points are isolated, must be closed?
SEE NOTE 74

4.3.2 If a set has no isolated points must it be closed? Must it be open?

4.3.3 A careless student, when asked, incorrectly remembers that a set is closed "if all its points are points of accumulation." Must such a set be closed?

4.3.4 A careless student, when asked, incorrectly remembers that a set is open "if it contains all of its interior points." Is there an example of a set that fails to have this property? Is there an example of a nonopen set that has this property?

4.3.5 Determine which of the following sets are open, which are closed, and which are neither open nor closed.

(a) $(-\infty,0)\cup(0,\infty)$

(b) $\{1,1/2,1/3,1/4,1/5,\dots\}$

(c) $\{0\}\cup\{1,1/2,1/3,1/4,1/5,\dots\}$

(d) $(0,1)\cup(1,2)\cup(2,3)\cup(3,4)\cup\cdots\cup(n,n+1)\cup\dots$

(e) $(1/2,1)\cup(1/4,1/2)\cup(1/8,1/4)\cup(1/16,1/8)\cup\dots$

(f) $\{x : |x-\pi| < 1\}$

(g) $\{x : x^2 < 2\}$

(h) $\mathbb{R}\setminus\mathbb{N}$

(i) $\mathbb{R}\setminus\mathbb{Q}$

4.3.6 Show that the closure operation has the following properties:

(a) If $E_1 \subset E_2$, then $\overline{E_1} \subset \overline{E_2}$.

(b) $\overline{E_1 \cup E_2} = \overline{E_1} \cup \overline{E_2}$.

(c) $\overline{E_1 \cap E_2} \subset \overline{E_1} \cap \overline{E_2}$.

(d) Give an example of two sets E_1 and E_2 such that
$$\overline{E_1 \cap E_2} \neq \overline{E_1} \cap \overline{E_2}.$$

(e) $\overline{\overline{E}} = \overline{E}$.

4.3.7 Show that the interior operation has the following properties:

(a) If $E_1 \subset E_2$, then $\text{int}(E_1) \subset \text{int}(E_2)$.

(b) $\text{int}(E_1 \cap E_2) = \text{int}(E_1) \cap \text{int}(E_2)$.

(c) $\text{int}(E_1 \cup E_2) \supset \text{int}(E_1) \cup \text{int}(E_2)$.

(d) Give an example of two sets E_1 and E_2 such that

$$\text{int}(E_1 \cup E_2) \neq \text{int}(E_1) \cup \text{int}(E_2).$$

(e) $\text{int}(\text{int}(E)) = \text{int}(E)$.

4.3.8 Show that if the set E' of points of accumulation of E is empty, then the set E must be closed.

4.3.9 Show that the set E' of points of accumulation of any set E must be closed.

4.3.10 Show that the set $\text{int}(E)$ of interior points of any set E must be open.

4.3.11 Show that a set E is closed if and only if $\overline{E} = E$.

4.3.12 Show that a set E is open if and only if $\text{int}(E) = E$.

4.3.13 If A is open and B is closed, what can you say about the sets $A \setminus B$ and $B \setminus A$?

4.3.14 If A and B are both open or both closed, what can you say about the sets $A \setminus B$ and $B \setminus A$?

4.3.15 If E is a nonempty bounded, closed set, show that $\max\{E\}$ and $\min\{E\}$ both exist. If E is a bounded, open set, show that neither $\max\{E\}$ nor $\min\{E\}$ exist (although $\sup\{E\}$ and $\inf\{E\}$ do).

4.3.16 Show that if a set of real numbers E has at least one point of accumulation, then for every $\varepsilon > 0$ there exist points $x, y \in E$ so that $0 < |x - y| < \varepsilon$.

4.3.17 Construct an example of a set of real numbers E that has no points of accumulation and yet has the property that for every $\varepsilon > 0$ there exist points $x, y \in E$ so that $0 < |x - y| < \varepsilon$.

4.3.18 Let $\{x_n\}$ be a sequence of real numbers. Let E denote the set of all numbers z that have the property that there exists a subsequence $\{x_{n_k}\}$ convergent to z. Show that E is closed.

4.3.19 Determine the components of the open set $\mathbb{R} \setminus \mathbb{N}$.

4.3.20 Let $F = \{0\} \cup \{1, 1/2, 1/3, 1/4, 1/5, \ldots\}$. Show that F is closed and determine the components of the open set $\mathbb{R} \setminus F$.

4.3.21 In the proof of Theorem 4.15 show that if x and y belong to the same component, then $I_x = I_y$, while if x and y do not belong to the same component, then I_x and I_y have no points in common.

4.3.22 In the proof of Theorem 4.15, after obtaining the collection of components $\{I_x : x \in G\}$, there remained the task of listing them. In classroom discussions the following suggestions were made as to how the components might be listed:

(a) List the components from largest to smallest.

(b) List the components from smallest to largest.

(c) List the components from left to right.

(d) List the components from right to left.

For each of these give an example of an open set with infinitely many components for which this strategy would work and also an example where it would fail.

4.3.23 In searching for interesting examples of open sets, you may have run out of ideas. Here is an example of a construction due to Cantor that has become the source for many important examples in analysis. We describe the component intervals of an open set G inside the interval $(0,1)$. At each "stage" n we shall describe 2^{n-1} components.

At the first stage, stage 1, take $(1/3,2/3)$ and at stage 2 take $(1/9,2/9)$ and $(7/9,8/9)$ and so on so that at each stage we take all the middle third intervals of the intervals remaining inside $(0,1)$. The set G is the open subset of $(0,1)$ having these intervals as components.

(a) Describe exactly the collection of intervals forming the components of G.

(b) What are the endpoints of the components. How do they relate to ternary expansions of numbers in $[0,1]$?

(c) What is the sum of the lengths of all components?

(d) Sketch a picture of the set G by illustrating the components at the first three stages.

(e) Show that if $x, y \in G$, $x < y$, but x and y are not in the same component, then there are infinitely many components of G in the interval (x,y).

See Note 75

4.4 Elementary Topology

The study of open and closed sets in any space is called *topology*. Our goal now is to find relations between these ideas and examine the properties of these sets. Much of this is a useful introduction to topology in any space; some is very specific to the real line, where the topological ideas are easier to sort out.

The first theorem establishes the connection between the open sets and the closed sets. They are not quite opposites. They are better described as "complementary."

Theorem 4.16 (Open vs. Closed) *Let A be a set of real numbers and $B = \mathbb{R} \setminus A$ its complement. Then A is open if and only if B is closed.*

Proof. If A is open and B fails to be closed then there is a point z that is a point of accumulation of B and yet is not in B. Thus z must be in A. But if z is a point in an open set it must be an interior point. Hence there is an interval $(z-\delta, z+\delta)$ contained entirely in A; such an interval contains no points of B. Hence z cannot be a point of accumulation of B. This is a contradiction and so we have proved that B must be closed if A is open.

Conversely, if B is closed and A fails to be open, then there is a point $z \in A$ that is not an interior point of A. Hence every interval $(z-\delta, z+\delta)$ must contain points outside of A, namely points in B. By definition this means that z is a point of accumulation of B. But B is closed and so z, which is a point in A, should really belong to B. This is a contradiction and so we have proved that A must be open if B is closed. ■

Theorem 4.17 (Properties of Open Sets) *Open sets of real numbers have the following properties:*

1. *The sets $\emptyset$ and $\mathbb{R}$ are open.*
2. *Any intersection of a finite number of open sets is open.*
3. *Any union of an arbitrary collection of open sets is open.*
4. *The complement of an open set is closed.*

Proof. The first assertion is immediate and the last we have already proved. The third is easy. Thus it is enough for us to prove the second assertion. Let us suppose that E_1 and E_2 are open. To show that $E_1 \cap E_2$ is also open we need to show that every point is an interior point. Let $z \in E_1 \cap E_2$. Then, since z is in both of the sets E_1 and E_2 and both are open there are intervals

$$(z-\delta_1, z+\delta_1) \subset E_1$$

and

$$(z-\delta_2, z+\delta_2) \subset E_2.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. We must then have

$$(z-\delta, z+\delta) \subset E_1 \cap E_2,$$

which shows that z is an interior point of $E_1 \cap E_2$. Since z is any point, this proves that $E_1 \cap E_2$ is open.

Having proved the theorem for two open sets, it now follows for three open sets since

$$E_1 \cap E_2 \cap E_3 = (E_1 \cap E_2) \cap E_3.$$

That any intersection of an arbitrary finite number of open sets is open now follows by induction. ■

Theorem 4.18 (Properties of Closed Sets) *Closed sets of real numbers have the following properties:*

1. *The sets $\emptyset$ and $\mathbb{R}$ are closed.*
2. *Any union of a finite number of closed sets is closed.*
3. *Any intersection of an arbitrary collection of closed sets is closed.*
4. *The complement of a closed set is open.*

Proof. Except for the second assertion these are easy or have already been proved. Let us prove the second one. Let us suppose that E_1 and E_2 are closed. To show that $E_1 \cup E_2$ is also closed we need to show that every accumulation point belongs to that set. Let z be an accumulation point of $E_1 \cup E_2$ that does not belong to the set. Since z is in neither of the closed sets E_1 and E_2, this point z cannot be an accumulation

point of either. Thus some interval $(z-\delta, z+\delta)$ contains no points of either E_1 or E_2. Consequently, that interval contains no points of $E_1 \cup E_2$ and z is not an accumulation point after all, contradicting our assumption. Since z is any accumulation point, this proves that $E_1 \cup E_2$ is closed.

Having proved the theorem for two closed sets, it now follows for three closed sets since

$$E_1 \cup E_2 \cup E_3 = (E_1 \cup E_2) \cup E_3.$$

That any union of an arbitrary finite number of closed sets is closed now follows by induction. ■

Exercises

4.4.1 Explain why it is that the sets $\emptyset$ and $\mathbb{R}$ are open and also closed.

4.4.2 Show that a union of an arbitrary collection of open sets is open.

4.4.3 Show that an intersection of an arbitrary collection of closed sets is closed.

4.4.4 Give an example of a sequence of open sets $G_1, G_2, G_3, \ldots$ whose intersection is neither open nor closed. Why does this not contradict Theorem 4.17?

4.4.5 Give an example of a sequence of closed sets $F_1, F_2, F_3, \ldots$ whose union is neither open nor closed. Why does this not contradict Theorem 4.18?

4.4.6 Show that the set $\overline{E}$ can be described as the *smallest closed set that contains every point of* E.

See Note 76

4.4.7 Show that the set $\text{int}(E)$ can be described as the *largest open set that is contained inside* E.

See Note 77

4.4.8 A function $f : \mathbb{R} \to \mathbb{R}$ is said to be *bounded at a point* x_0 provided that there are positive numbers ε and M so that $|f(x)| < M$ for all $x \in (x_0 - \varepsilon, x_0 + \varepsilon)$. Show that the set of points at which a function is bounded is open. Let E be an arbitrary closed set. Is it possible to construct a function $f : \mathbb{R} \to \mathbb{R}$ so that the set of points at which f is not bounded is precisely the set E?

4.4.9 This exercise continues Exercise 4.3.23. Define the *Cantor ternary set* K to be the complement of the open set G of Exercise 4.3.23 in the interval $[0,1]$.

(a) If all the open intervals up to the nth stage in the construction of G are removed from the interval $[0,1]$, there remains a closed set K_n that is the union of a finite number of closed intervals. How many intervals?

(b) What is the sum of the lengths of these closed intervals that make up K_n?

(c) Show that $K = \bigcap_{n=1}^{\infty} K_n$.

(d) Sketch a picture of the set K by illustrating the sets K_1, K_2, and K_3.

(e) Show that if $x, y \in K$, $x < y$, then there is an open subinterval $I \subset (x,y)$ containing no points of K.

(f) Give an example of a number $z \in K \cap (0,1)$ that is not an endpoint of a component of G.

4.4.10 Express the closed interval $[0,1]$ as an intersection of a sequence of open sets. Can it also be expressed as a union of a sequence of open sets?

4.4.11 Express the open interval $(0,1)$ as a union of a sequence of closed sets. Can it also be expressed as an intersection of a sequence of closed sets?

4.5 Compactness Arguments

✂ Parts of this section could be cut in a short course. For a minimal approach to compactness arguments, you may wish to skip over all but the Bolzano-Weierstrass property. For all purposes of elementary real analysis this is sufficient. Proofs in the sequel that require a compactness argument will be supplied with one that uses the Bolzano-Weierstrass property and, perhaps, another that can be omitted.

In analysis we frequently encounter the problem of arguing from a set of "local" assumptions to a "global" conclusion. Let us focus on just one problem of this type and see the kind of arguments that can be used.

Local Boundedness of a Function Suppose that a function f is *locally bounded* at each point of a set E. By this we mean that for every point $x \in E$ there is an interval $(x-\delta, x+\delta)$ and f is bounded on the points in E that belong to that interval. Can we conclude that f is bounded on the whole of the set E?

Thus we have been given a local condition at each point x in the set E. There must be numbers δ_x and M_x so that

$$|f(t)| \leq M_x \text{ for all } t \in E \text{ in the interval } (x-\delta_x, x+\delta_x).$$

The global condition we want, if possible, is to have some single number M that works for all $t \in E$; that is,

$$|f(t)| \leq M \text{ for all } t \in E.$$

Two examples show that this depends on the nature of the set E.

Example 4.19: The function $f(x) = 1/x$ is locally bounded at each point x in the set $(0,1)$ but is not bounded on the set $(0,1)$. It is clear that f cannot be bounded on $(0,1)$ since the statement

$$\frac{1}{t} \leq M \text{ for all } t \in (0,1)$$

cannot be true for any M. But this function is locally bounded at each point x here. Let $x \in (0,1)$. Take $\delta_x = x/2$ and $M_x = 2/x$. Then

$$f(t) = \frac{1}{t} \leq \frac{2}{x} = M_x$$

if

$$x/2 = x - \delta_x < t < x + \delta_x.$$

What is wrong here? What is there about this set $E = (0,1)$ that does not allow the conclusion? The point 0 is a point of accumulation of $(0,1)$ that does not belong to $(0,1)$, and so there is no assumption that f is bounded at that point. We avoid this difficulty if we assume that E is closed. ◀

Example 4.20: The function $f(x) = x$ is locally bounded at each point x in the set $[0,\infty)$ but is not bounded on the set $[0,\infty)$. It is clear that f cannot be bounded on $[0,\infty)$ since the statement

$$f(t) = t \leq M \text{ for all } t \in [0,\infty)$$

cannot be true for any M. But this function is locally bounded at each point x here. Let $x \in [0,\infty)$. Take $\delta_x = 1$ and $M_x = x+1$. Then

$$f(t) = t \leq x+1 = M_x$$

if $x-1 < t < x+1$.

What is wrong here? What is there about this set $E = [0,\infty)$ that does not allow the conclusion? This set is closed and so contains all of its accumulation points so that the difficulty we saw in the preceding example does not arise. The difficulty is that the set is too big, allowing larger and larger bounds as we move to the right. We could avoid this difficulty if we assume that E is bounded. ◀

Indeed, as we shall see, we have reached the correct hypotheses now for solving our problem. The version of the theorem we were searching for is this:

> **Theorem** *Suppose that a function f is locally bounded at each point of a closed and bounded set E. Then f is bounded on the whole of the set E.*

Arguments that exploit the special features of closed and bounded sets of real numbers are called *compactness arguments*. Most often they are used to prove that some local property has global implications, which is precisely the nature of our boundedness theorem. We now solve our problem using various different compactness arguments. Each of these arguments will become a formidable tool in proving theorems in analysis. Many situations will arise in which some local property must be proved to hold globally, and compactness will play a huge role in these.

4.5.1 Bolzano-Weierstrass Property

A closed and bounded set has a special feature that can be used to design compactness arguments. This property is essentially a repeat of a property about convergent subsequences that we saw in Section 2.11.

Theorem 4.21 (Bolzano-Weierstrass Property) *A set of real numbers E is closed and bounded if and only if every sequence of points chosen from the set has a subsequence that converges to a point that belongs to E.*

Proof. Suppose that E is both closed and bounded and let $\{x_n\}$ be a sequence of points chosen from E. Since E is bounded this sequence $\{x_n\}$ must be bounded too. We apply the Bolzano-Weierstrass theorem for sequences (Theorem 2.40) to obtain a subsequence $\{x_{n_k}\}$ that converges. If $x_{n_k} \to z$ then since all the points of the subsequence belong to E either the sequence is constant after some term or else z is a point of accumulation of E. In either case we see that $z \in E$. This proves the theorem in one direction.

In the opposite direction we suppose that a set E, which we do not know in advance to be either closed or bounded, has the Bolzano-Weierstrass property. Then E cannot be unbounded. For example, if E is unbounded then there is a sequence of points $\{x_n\}$ of E with $x_n \to \infty$ or $-\infty$ and no subsequence of that sequence converges, contradicting the assumption.

Also, E must be closed. If not, there is a point of accumulation z that is not in E. This means that there is a sequence of points $\{x_n\}$ in E converging to z. But any subsequence of $\{x_n\}$ would also converge to z and, since $z \notin E$, we again have a contradiction. ■

This theorem can also be interpreted as a statement about accumulation points.

Corollary 4.22: *A set of real numbers E is closed and bounded if and only if every infinite subset of E has a point of accumulation that belongs to E.*

Let us use the Bolzano-Weierstrass property to prove our theorem about local boundedness.

> **Theorem** *Suppose that a function f is locally bounded at each point of a closed and bounded set E. Then f is bounded on the whole of the set E.*

Proof. **(Bolzano-Weierstrass compactness argument)** To use this argument we will need to construct a sequence of points in E that we can use. Our proof is a proof by contradiction. If f is not bounded on E there must be a sequence of points $\{x_n\}$ chosen from E so that

$$|f(x_n)| > n$$

for all n. If such a sequence could not be chosen, then at some stage, N say, there are no more points with $|f(x_N)| > N$ and N is an upper bound.

By compactness (i.e., by Theorem 4.21) there is a convergent subsequence $\{x_{n_k}\}$ converging to a point $z \in E$. By the local boundedness assumption there is an open interval $(z-\delta, z+\delta)$ and a number M_z so that

$$|f(t)| \leq M_z$$

whenever t is in E and inside that interval. But for all sufficiently large values of k, the point x_{n_k} must belong to the interval $(z-\delta, z+\delta)$. The two statements

$$|f(x_{n_k})| > n_k \text{ and } |f(x_{n_k})| \leq M_z$$

cannot both be true for all large k and so we have reached a contradiction, proving the theorem. ■

4.5.2 Cantor's Intersection Property

✂ *Enrichment section. May be omitted.*

A famous compactness argument, one that is used often in analysis, involves the intersection of a *descending* sequence of sets; that is, a sequence with

$$E_1 \supset E_2 \supset E_3 \supset E_4 \supset \ldots.$$

What conditions on the sequence will imply that

$$\bigcap_{n=1}^{\infty} E_n \neq \emptyset?$$

Example 4.23: An example shows that some conditions are needed. Suppose that for each $n \in \mathbb{N}$ we let $E_n = (0, 1/n)$. Then

$$E_1 \supset E_2 \supset E_3 \supset \ldots,$$

so $\{E_n\}$ is a descending sequence of sets with empty intersection. The same is true of the sequence $F_n = [n, \infty)$. Observe that the sets in the sequence $\{E_n\}$ are bounded (but not closed) while the sets in the sequence $\{F_n\}$ are closed (but not bounded). ◀

In a paper in 1879 Cantor described the following theorem and the role it plays in analysis. He pointed out that variants on this idea had been already used throughout most of that century, notably by Lagrange, Legendre, Dirichlet, Cauchy, Bolzano, and Weierstrass.

Theorem 4.24: *Let $\{E_n\}$ be a sequence of nonempty closed and bounded subsets of real numbers such that $E_1 \supset E_2 \supset E_3 \supset \ldots$. Let $E = \bigcap_{n=1}^{\infty} E_n$. Then E is not empty.*

Proof. For each $i \in \mathbb{N}$ choose $x_i \in E_i$. The sequence $\{x_i\}$ is bounded since every point lies inside the bounded set E_1. Therefore, because of Theorem 4.21, $\{x_i\}$ has a convergent subsequence $\{x_{i_k}\}$. Let z denote that limit. Fix an integer m. Because the sets are descending, $x_{i_k} \in E_m$ for all sufficiently large $k \in \mathbb{N}$. But E_m is closed, from which it follows that $z \in E_m$. This is true for all $m \in \mathbb{N}$, so $z \in E$. ■

Corollary 4.25 (Cantor Intersection Theorem) *Suppose that $\{E_n\}$ is a sequence of nonempty closed subsets of real numbers such that*

$$E_1 \supset E_2 \supset E_3 \supset \ldots.$$

If

$$\text{diameter } E_n \to 0,$$

then the intersection

$$E = \bigcap_{n=1}^{\infty} E_n$$

consists of a single point.

Proof. Here the diameter of a nonempty, closed bounded set E would just be $\max E - \min E$, which exists and is finite for such a set (see Exercise 4.3.15). Since we are assuming that the diameters shrink to zero it follows that, at least for all sufficiently large n, E_n must be bounded.

That $E \neq \emptyset$ follows from Theorem 4.24. It remains to show that E contains only one point. Let $x \in E$ and $y \in \mathbb{R}$, $y \neq x$. Since diameter $E_n \to 0$, there exists $i \in \mathbb{N}$ such that diameter $E_i < |x - y|$. Since $x \in E_i$, y cannot be in E_i. Thus $y \notin E$ and $E = \{x\}$ as required. ■

Now we prove our theorem about local boundedness by using the Cantor intersection property to frame an argument.

Theorem *Suppose that a function f is locally bounded at each point of a closed and bounded set E. Then f is bounded on the whole of the set E.*

Proof. **(Cantor intersection compactness argument).** To use this argument we will need to construct a sequence of closed and bounded sets shrinking to a point. Our proof is again a proof by contradiction. Suppose that f is not bounded on E.

Since E is bounded we may assume that E is contained in some interval $[a,b]$. Divide that interval in half, forming two subintervals of the same length, namely $(b-a)/2$. At least one of these intervals contains points of E and f is unbounded on that interval. Call it $[a_1,b_1]$.

Now do the same to the interval $[a_1,b_1]$. Divide that interval in half, forming two subintervals of the same length, namely $(b-a)/4$. At least one of these intervals contains points of E and f is unbounded on that interval. Call it $[a_2,b_2]$. Continue this process inductively, producing a descending sequence of intervals $\{[a_n,b_n]\}$ so that the nth interval $[a_n,b_n]$ has length $(b-a)/2^n$, contains points of E, and f is unbounded on $E\cap[a_n,b_n]$.

By the Cantor intersection property there is a single point $z\in E$ contained in all of these intervals. But by our local boundedness assumption there is an interval $(z-c,z+c)$ so that f is bounded on the points of E in that interval. For any large enough value of n, though, the interval $[a_n,b_n]$ would be contained inside the interval $(z-c,z+c)$. This would be impossible and so we have reached a contradiction, proving the theorem. ■

4.5.3 Cousin's Property

✂ *Enrichment section. May be omitted.*

Another compactness argument dates back to Pierre Cousin in the last years of the nineteenth century. This exploits the order of the real line and considers how small intervals may be pieced together to give larger intervals. The larger interval $[a,b]$ is subdivided

$$a=x_0<x_1<\cdots<x_n=b$$

and then expressed as a finite union of nonoverlapping subintervals said to form a *partition*:

$$[a,b]=\bigcup_{i=1}^{n}[x_{i-1},x_i].$$

This again provides us with a compactness argument since it allows a way to argue from the local to the global.

Lemma 4.26 (Cousin) *Let $\mathcal{C}$ be a collection of closed subintervals of $[a,b]$ with the property that for each $x\in[a,b]$ there exists $\delta=\delta(x)>0$ such that $\mathcal{C}$ contains all intervals $[c,d]\subset[a,b]$ that contain x and have length smaller than δ. Then there exists a partition*

$$a=x_0<x_1<\cdots<x_n=b$$

of $[a,b]$ such that $[x_{i-1},x_i]\in\mathcal{C}$ for $i=1,\ldots,n$.

This lemma makes precise the statement that if a collection of closed intervals contains all "sufficiently small" ones for $[a,b]$, then it contains a partition of $[a,b]$. We shall frequently see the usefulness of such a partition. This is the most elementary of a collection of tools called *covering theorems*. Roughly, a *cover* of a set is a family of intervals covering the set in the sense that each point in the set is contained in one or more of the intervals.

We formalize the assumption in Cousin's lemma in this language:

Definition 4.27: (Cousin Cover) A collection $\mathcal{C}$ of closed intervals satisfying the hypothesis of Cousin's lemma is called a *Cousin cover* of $[a,b]$.

Proof. **(Proof of Cousin's lemma)** Let us, in order to obtain a contradiction, suppose that $\mathcal{C}$ does not contain a partition of the interval $[a,b]$. Let c be the midpoint of that interval and consider the two subintervals $[a,c]$ and $[c,b]$. If $\mathcal{C}$ contains a partition of both intervals $[a,c]$ and $[c,b]$, then by putting those partitions together we can obtain a partition of $[a,b]$, which we have supposed is impossible.

Let $I_1 = [a,b]$ and let I_2 be either $[a,c]$ or $[c,b]$ chosen so that $\mathcal{C}$ contains no partition of I_2. Inductively we can continue in this fashion, obtaining a shrinking sequence of intervals $I_1 \supset I_2 \supset I_3 \supset \dots$ so that the length of I_n is $(b-a)/2^{n-1}$ and $\mathcal{C}$ contains no partition of I_n.

By the Cantor intersection theorem (Theorem 4.25) there is a single point z in all of these intervals.

For sufficiently large n, the interval I_n contains z and has length smaller than $\delta(z)$. Thus, by definition, $I_n \in \mathcal{C}$. In particular, $\mathcal{C}$ does indeed contain a partition of that interval I_n since the single interval $\{I_n\}$ is itself a partition. But this contradicts the way in which the sequence was chosen and this contradiction completes our proof. ■

Now we reprove our theorem about local boundedness by using Cousin's property to frame an argument.

> **Theorem** *Suppose that a function f is locally bounded at each point of a closed and bounded set E. Then f is bounded on the whole of the set E.*

Proof. **(Cousin compactness argument)** The set E is bounded and so is contained in some interval $[a,b]$. Let us say that an interval $[c,d] \subset [a,b]$ is "black" if the following statement is true:

> There is a number M (which may depend on $[c,d]$) so that $|f(t)| \leq M$ for all $t \in E$ that are in the interval $[c,d]$.

The collection of all black intervals is a Cousin cover of $[a,b]$. This is because of the local boundedness assumption on f. Consequently, by Cousin's lemma, there is a partition of the interval $[a,b]$ consisting of black intervals. The function f is bounded in E on each of these finitely many black intervals and so, since there are only finitely many of them, f must be bounded on E in $[a,b]$. But $[a,b]$ includes all of E and so the proof is complete. ■

4.5.4 Heine-Borel Property

✂ *Advanced section. May be omitted.*

Another famous compactness property involves covers too, as in the Cousin lemma, but this time covers consisting of open intervals. This theorem has wide applications, including again extensions of local properties to global ones. You may find this compactness argument more difficult to work with than the others. On the real line all of the arguments here are equivalent and, in most cases, any one will do the job. Why not use the simpler ones then? The answer is that in more general spaces than the real line these other versions may be more useful. Time spent learning them now will pay off in later courses.

The property we investigate is named after two mathematicians, Émile Borel (1871–1956) and Heinrich Eduard Heine (1821–1881), whose names have become closely attached to these ideas.

We begin with some definitions.

Definition 4.28: (Open Cover) Let $A \subset \mathbb{R}$ and let $\mathcal{U}$ be a family of open intervals. If for every $x \in A$ there exists at least one interval $U \in \mathcal{U}$ such that $x \in U$, then $\mathcal{U}$ is called an *open cover* of A.

Definition 4.29: (Heine-Borel Property) A set $A \subset \mathbb{R}$ is said to have the *Heine-Borel property* if every open cover of A can be reduced to a finite subcover. That is, if $\mathcal{U}$ is an open cover of A, then there exists a finite subset of $\mathcal{U}$, $\{U_1, U_2, \ldots, U_n\}$ such that

$$A \subset U_1 \cup U_2 \cup \cdots \cup U_n.$$

Example 4.30: Any finite set has the Heine-Borel property. Just take one interval from the cover for each element in the finite set. ◀

Example 4.31: The set $\mathbb{N}$ does not have the Heine-Borel property. Take, for example, the collection of open intervals

$$\{(0,n) : n = 1,2,3,\ldots\}.$$

While this forms an open cover of $\mathbb{N}$, no finite subcollection could also be an open cover. ◀

Example 4.32: The set $A = \{1/n : n \in \mathbb{N}\}$ does not have the Heine-Borel property. Take, for example, the collection of open intervals

$$\{(1/n,2) : n = 1,2,3,\ldots\}.$$

While this forms an open cover of A, no finite subcollection could also be an open cover. ◀

Observe in these examples that $\mathbb{N}$ is closed (but not bounded) while A is bounded (but not closed). We shall prove, in Theorem 4.33, that a set A has the Heine-Borel property if and only if that set is both closed and bounded.

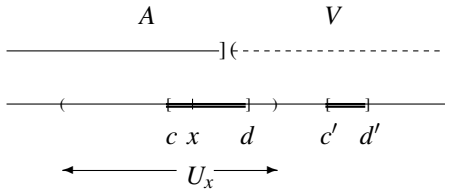


Figure 4.2. The two types of intervals in the proof of Theorem 4.33.

Theorem 4.33 (Heine-Borel) *A set $A \subset \mathbb{R}$ has the Heine-Borel property if and only if A is both closed and bounded.*

Proof. Suppose $A \subset \mathbb{R}$ is both closed and bounded, and $\mathcal{U}$ is an open cover for A. We may assume $A \neq \emptyset$, otherwise there is nothing to prove. Let $[a,b]$ be the smallest closed interval containing A; that is,

$$a = \inf\{x : x \in A\} \text{ and } b = \sup\{x : x \in A\}.$$

Observe that $a \in A$ and $b \in A$. We shall apply Cousin's lemma to the interval $[a,b]$, so we need to first define an appropriate Cousin cover of $[a,b]$.

For each $x \in A$, since $\mathcal{U}$ is an open cover of A, there exists an open interval $U_x \in \mathcal{U}$ such that $x \in U_x$. Since U_x is open, there exists $\delta(x) > 0$ for which $(x-t, x+t) \subset U_x$ for all $t \in (0, \delta(x))$. This defines $\delta(x)$ for points in A. Now consider points in $V = [a,b] \setminus A$. We must define $\delta(x)$ for points of V. Since A is closed and $\{a,b\} \subset A$, V is open (why?); thus for each $x \in V$ there exists $\delta(x) > 0$ such that $(x-t, x+t) \subset V$ for all $t \in (0, \delta(x))$. We can therefore obtain a Cousin cover $\mathcal{C}$ of $[a,b]$ as follows: An interval $[c,d]$ is a member of $\mathcal{C}$ if there exists $x \in [a,b]$ such that either (i) $x \in A$ and $x \in [c,d] \subset U_x$ or (ii) $x \in V$ and $x \in [c,d] \subset V$.

Observe that an interval of type (i) can contain points of V, but an interval of type (ii) cannot contain points of A. Figure 4.2 illustrates examples of both types of intervals. In that figure $[c,d] \subset U_x$ is an interval of type (i) in $\mathcal{C}$; $[c',d'] \subset V$ is an interval of type (ii) in $\mathcal{C}$.

It is clear that $\mathcal{C}$ forms a Cousin cover of $[a,b]$. From Cousin's lemma we infer the existence of a partition $a = x_0 < x_1 < \cdots < x_n = b$ with $[x_{i-1}, x_i] \in \mathcal{C}$ for $i = 1, \ldots, n$. Each of the intervals $[x_{i-1}, x_i]$ is either contained in V (in which case it is disjoint from A) or is contained in some member $U_i \in \mathcal{U}$. We now "throw away" from the partition those intervals that contain only points of V, and the union of the remaining closed intervals covers all of A. Each interval of this finite collection is contained in some open interval U from the cover $\mathcal{U}$. More precisely, let

$$S = \{i : 1 \leq i \leq n \text{ and } [x_{i-1}, x_i] \subset U_i\}.$$

Then

$$A \subset \bigcup_{i \in S} [x_{i-1}, x_i] \subset \bigcup_{i \in S} U_i,$$

so

$$\{U_i : i \in S\}$$

is the required subcover of A.

To prove the converse, we must show that if A is not bounded or if A is not closed, then there exists an open cover of A with no finite subcover. Suppose first that A is not bounded. Consider the family of open intervals

$$\mathcal{U} = \{(-n,n) : n \in \mathbb{N}\}.$$

Clearly $\mathcal{U}$ is an open cover of A. (Indeed it is an open cover of all of $\mathbb{R}$.) But it is also clear that $\mathcal{U}$ contains no finite subcover of A since a finite subcover will cover only a bounded set and we have assumed that A is unbounded.

Now suppose A is not closed. Then there is a point of accumulation z of A that does not belong to A. Consider the family of open intervals

$$\mathcal{U} = \left\{\left(-\infty, z - \frac{1}{n}\right) : n \in \mathbb{N}\right\} \cup \left\{\left(z + \frac{1}{n}, \infty\right) : n \in \mathbb{N}\right\}.$$

Clearly $\mathcal{U}$ is an open cover of A. (Indeed it is an open cover of all of $\mathbb{R} \setminus \{z\}$.) But it is also clear that $\mathcal{U}$ contains no finite subcover of A since a finite subcover contains no points of the interval $(z-c, z+c)$ for some small positive c and yet, since z is an accumulation point of A, this interval must contain infinitely many points of A. ■

Once again, we return to our sample theorem, which shows how a local property can be used to prove a global condition, this time using a Heine-Borel compactness argument.

> **Theorem** *Suppose that a function f is locally bounded at each point of a closed and bounded set E. Then f is bounded on the whole of the set E.*

Proof. **(Heine-Borel compactness argument).** As f is locally bounded at each point of E, for every $x \in E$ there exists an open interval U_x containing x and a positive number M_x such that $|f(t)| < M_x$ for all $t \in U_x \cap E$. Let

$$\mathcal{U} = \{U_x : x \in E\}.$$

Then $\mathcal{U}$ is an open cover of E. By the Heine-Borel theorem there exists

$$\{U_{x_1}, U_{x_2}, \ldots, U_{x_n}\}$$

such that

$$E \subset U_{x_1} \cup U_{x_2} \cup \cdots \cup U_{x_n}.$$

Let

$$M = \max\{M_{x_1}, M_{x_2}, \ldots, M_{x_n}\}.$$

Let $x \in E$. Then there exists i, $1 \leq i \leq n$, for which $x \in U_{x_i}$. Since

$$|f(x)| \leq M_{x_i} \leq M$$

we conclude that f is bounded on E. ■

Our ability to reduce $\mathcal{U}$ to a *finite* subcover in the proof of this theorem was crucial. You may wish to use the function $f(x) = 1/x$ on $(0,1]$ to appreciate this statement.

4.5.5 Compact Sets

We have seen now a wide range of techniques called compactness arguments that can be applied to a set that is closed and bounded. We now introduce the modern terminology for such sets.

Definition 4.34: A set of real numbers E is said to be *compact* if it has any of the following equivalent properties:

1. E is closed and bounded.
2. E has the Bolzano-Weierstrass property.
3. E has the Heine-Borel property.

In spaces more general than the real line there may be analogues of the notions of closed, bounded, convergent sequences, and open covers. Thus there can also be analogues of closed and bounded sets, the Bolzano-Weierstrass property, and the Heine-Borel property. In these more general spaces the three properties are not always equivalent and it is the Heine-Borel property that is normally chosen as the definition of compact sets there. Even so, a thorough understanding of compactness arguments on the real line is an excellent introduction to these advanced and important ideas in other settings.

If we return to our sample theorem we see that now, perhaps, it should best be described in the language of compact sets:

> **Theorem** *Suppose that E is compact. Then every function $f : E \to \mathbb{R}$ that is locally bounded on E is bounded on the whole of the set E. Conversely, if every function $f : E \to \mathbb{R}$ that is locally bounded on E is bounded on the whole of the set E, then E must be compact.*

In real analysis there are many theorems of this type. The concept of compact set captures exactly when many local conditions can have global implications.

Exercises

4.5.1 Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ that is not locally bounded at any point.
SEE NOTE 78

4.5.2 Show directly that the interval $[0, \infty)$ does not have the Bolzano-Weierstrass property.

4.5.3 ✂ Show directly that the interval $[0, \infty)$ does not have the Heine-Borel property.

4.5.4 ✂ Show directly that the set $[0,1] \cap \mathbb{Q}$ does not have the Heine-Borel property.

4.5.5 Develop the properties of compact sets: (i) is the union of a pair of compact sets compact? (ii) is the union of a finite sequence of compact sets compact? (iii) is the union of a sequence of compact sets compact? Do the same for intersections.
SEE NOTE 79

4.5.6 Show directly that the union of two sets with the Bolzano-Weierstrass property must have the Bolzano-Weierstrass property.

4.5.7 ✂ Show directly that the union of two sets with the Heine-Borel property must have the Heine-Borel property.

4.5.8 We defined an open cover of a set E to consist of open *intervals* covering E. Let us change that definition to allow an open cover to consist of any family of open *sets* covering E. What changes are needed in the proof of Theorem 4.33 so that it remains valid in this greater generality?

SEE NOTE 80

4.5.9 A function $f : \mathbb{R} \to \mathbb{R}$ is said to be *locally increasing* at a point x_0 if there is a $\delta > 0$ so that

$$f(x) < f(x_0) < f(y)$$

whenever

$$x_0 - \delta < x < x_0 < y < x_0 + \delta.$$

Show that a function that is locally increasing at every point in $\mathbb{R}$ must be increasing; that is, that $f(x) < f(y)$ for all $x < y$.

SEE NOTE 81

4.5.10 Let $f : E \to \mathbb{R}$ have this property: For every $e \in E$ there is an $\varepsilon > 0$ so that

$$f(x) > \varepsilon \text{ if } x \in E \cap (e - \varepsilon, e + \varepsilon).$$

Show that if the set E is compact then there is some positive number c so that

$$f(e) > c$$

for all $e \in E$. Show that if E is not closed or is not bounded, then this conclusion may not be valid.

4.5.11 Prove the following variant of Lemma 4.26:

> Let $\mathcal{C}$ be a collection of closed subintervals of $[a,b]$ with the property that for each $x \in [a,b]$ there exists $\delta = \delta(x) > 0$ such that $\mathcal{C}$ contains all intervals $[c,d] \subset [a,b]$ that contain x and have length smaller than δ. Suppose that $\mathcal{C}$ has the property that if $[\alpha,\beta]$ and $[\beta,\gamma]$ both belong to $\mathcal{C}$ then so too does $[\alpha,\gamma]$. Then $[a,b]$ belongs to $\mathcal{C}$.

4.5.12 Use the version of Cousin's lemma given in Exercise 4.5.11 to give a simpler proof of the sample theorem on local boundedness.

4.5.13 ✂ Give an example of an open covering of the set $\mathbb{Q}$ of rational numbers that does not reduce to a finite subcover.

4.5.14 Suppose that E is closed and K is compact. Show that $E \cap K$ is compact. Do this in two ways (using the "closed + bounded" definition and also using the Bolzano-Weierstrass property).

4.5.15 Prove that every function $f : E \to \mathbb{R}$ that is locally bounded on E is bounded on the whole of the set E only if the set E is compact, by supplying the following two constructions:

(a) Show that if the set E is not bounded, then there is an unbounded function $f : E \to \mathbb{R}$ so that f is locally bounded on E.

(b) Show that if the set E is not closed, then there is an unbounded function $f : E \to \mathbb{R}$ so that f is locally bounded on E.

4.5.16 ✂ Suppose that E is closed and K is compact. Show that $E \cap K$ is compact using the Heine-Borel property.

4.5.17 Suppose that E is compact. Is the set of boundary points of E also compact?

4.5.18 ✂ Prove Lindelöff's covering theorem:

> *Let $\mathcal{C}$ be a collection of open intervals such that every point of a set E belongs to at least one of the intervals. Then there is a sequence of intervals $I_1, I_2, I_3, \ldots$ chosen from $\mathcal{C}$ that also covers E.*

SEE NOTE 82

4.5.19 Describe briefly the distinction between the covering theorem of Lindelöff (Exercise 4.5.18) and that of Heine-Borel.
SEE NOTE 83

4.5.20 ✂ We have seen that the following four conditions on a set $A \subset \mathbb{R}$ are equivalent:

(a) A is closed and bounded.

(b) Every infinite subset of A has a limit point in A.

(c) Every sequence of points from A has a subsequence converging to a point in A.

(d) Every open cover of A has a finite subcover.

Prove directly that (b)⇒(c), (b)⇒(d) and (c)⇒(d).
SEE NOTE 84

4.5.21 Let f be a function that is locally bounded on a compact interval $[a,b]$. Let

$$S = \{a < x \leq b : \ f \text{ is bounded on } [a,x]\}.$$

(a) Show that $S \neq \emptyset$.

(b) Show that if $z = \sup S$, then $a < z \leq b$.

(c) Show that $z \in S$.

(d) Show that $z = b$ by showing that $z < b$ is impossible.

Using these steps, construct a proof of the sample theorem on local boundedness.

4.6 Countable Sets

As part of our discussion of properties of sets in this chapter let us review a special property of sets that relates, not to their topological properties, but to their size. We can divide sets into finite sets and infinite sets. How do we divide infinite sets into "large" and "larger" infinite sets?

We did this in our discussion of sequences in Section 2.3. (If you skipped over that section now is a good time to go back.) If an infinite set E has the property that the elements of E can be written as a list (i.e., as a sequence)

$$e_1, e_2, e_3, \ldots, e_n, \ldots,$$

then that set is said to be *countable*. Note that this property has nothing particularly to do with the other properties of sets encountered in this chapter. It is yet another and different way of classifying sets.

The following properties review our understanding of countable sets. Remember that the empty set, any finite set, and any infinite set that can be listed are all said to be countable. An infinite set that cannot be listed is said to be *uncountable*.

Theorem 4.35: *Countable sets have the following properties:*

1. *Any subset of a countable set is countable.*
2. *Any union of a sequence of countable sets is countable.*
3. *No interval is countable.*

Exercises

4.6.1 Give examples of closed sets that are countable and closed sets that are uncountable.

4.6.2 Is there a nonempty open set that is countable?

4.6.3 If a set is countable, what can you say about its complement?

4.6.4 Is the intersection of two uncountable sets uncountable?

4.6.5 Show that the Cantor set of Exercise 4.3.23 is infinite and uncountable.
SEE NOTE 85

4.6.6 Give (if possible) an example of a set with

(a) Countably many points of accumulation
(b) Uncountably many points of accumulation
(c) Countably many boundary points
(d) Uncountably many boundary points
(e) Countably many interior points
(f) Uncountably many interior points

4.6.7 A set is said to be *co-countable* if it has a countable complement. Show that the intersection of finitely many co-countable sets is itself co-countable.

4.6.8 Let E be a set and $f : \mathbb{R} \to \mathbb{R}$ be an *increasing* function [i.e., if $x < y$, then $f(x) < f(y)$]. Show that E is countable if and only if the image set $f(E)$ is countable. (What property other than "increasing" would work here?)

4.6.9 Show that every uncountable set of real numbers has a point of accumulation.
SEE NOTE 86

4.6.10 Let $\mathcal{F}$ be a family of (nondegenerate) intervals; that is, each member of $\mathcal{F}$ is an interval (open, closed or neither) but is not a single point. Suppose that any two intervals I and J in the family have no point in common. Show that the family $\mathcal{F}$ can be arranged in a sequence $I_1, I_2, \ldots$.
SEE NOTE 87

4.7 Challenging Problems for Chapter 4

4.7.1 Cantor, in 1885, defined a set E to be *dense-in-itself* if $E \subset E'$. Develop some facts about such sets. Include illustrative examples.

4.7.2 One of Cantor's early results in set theory is that for every closed set E there is a set S with $E = S'$. Attempt a proof.

4.7.3 Can the closed interval $[0,1]$ be expressed as the union of a sequence of disjoint closed subintervals each of length smaller than 1?

4.7.4 In many applications of open sets and closed sets we wish to work just inside some other set A. It is convenient to have a language for this. A set $E \subset A$ is said to be *open relative to* A if $E = A \cap G$ for some set $G \subset \mathbb{R}$ that is open. A set $E \subset A$ is said to be *closed relative to* A if $E = A \cap F$ for some set $F \subset \mathbb{R}$ that is closed. Answer the following questions.

(a) Let $A = [0,1]$ describe, if possible, sets that are open relative to A but not open as subsets of $\mathbb{R}$.

(b) Let $A = [0,1]$ describe, if possible, sets that are closed relative to A but not closed as subsets of $\mathbb{R}$.

(c) Let $A = (0,1)$ describe, if possible, sets that are open relative to A but not open as subsets of $\mathbb{R}$.

(d) Let $A = (0,1)$ describe, if possible, sets that are closed relative to A but not closed as subsets of $\mathbb{R}$.

4.7.5 Let $A = \mathbb{Q}$. Give examples of sets that are neither open nor closed but are both relative to $\mathbb{Q}$.

4.7.6 Show that all the subsets of $\mathbb{N}$ are both open and closed relative to $\mathbb{N}$.

4.7.7 Introduce for any set $E \subset \mathbb{R}$ the notation

$$\partial E = \{x : \ x \text{ is a boundary point of } E\}.$$

(a) Show for any set E that $\partial E = \overline{E} \cap \overline{(\mathbb{R} \setminus E)}$.

(b) Show that for any set E the set ∂E is closed.

(c) For what sets E is it true that $\partial E = \emptyset$?

(d) Show that $\partial E \subset E$ for any closed set E.

(e) If E is closed, show that $\partial E = E$ if and only if E has no interior points.

(f) If E is open, show that ∂E can contain no interval.

4.7.8 Let E be a nonempty set of real numbers and define the function

$$f(x) = \inf\{|x - e| : e \in E\}.$$

(a) Show that $f(x) = 0$ for all $x \in E$.

(b) Show that $f(x) = 0$ if and only if $x \in \overline{E}$.

(c) Show for any nonempty closed set E that

$$\{x \in \mathbb{R} : f(x) > 0\} = (\mathbb{R} \setminus E).$$

4.7.9 Let $f : \mathbb{R} \to \mathbb{R}$ have this property: For every $x_0 \in \mathbb{R}$ there is a $\delta > 0$ so that

$$|f(x) - f(x_0)| < |x - x_0|$$

whenever $0 < |x - x_0| < \delta$. Show that for all $x, y \in \mathbb{R}$, $x \neq y$,

$$|f(x) - f(y)| < |x - y|.$$

4.7.10 Let $f : E \to \mathbb{R}$ have this property: For every $e \in E$ there is an $\varepsilon > 0$ so that

$$f(x) > \varepsilon \text{ if } x \in E \cap (e - \varepsilon, e + \varepsilon).$$

Show that if the set E is compact, then there is some positive number c so that

$$f(e) > c$$

for all $e \in E$. Show that if E is not closed or is not bounded, then this conclusion may not be valid.

4.7.11 **(Separation of Compact Sets)** Let A and B be nonempty sets of real numbers and let

$$\delta(A, B) = \inf\{|a - b| : a \in A, b \in B\}.$$

$\delta(A, B)$ is often called the "distance" between the sets A and B.

(a) Prove $\delta(A, B) = 0$ if $A \cap B \neq \emptyset$.

(b) Give an example of two closed, disjoint sets in $\mathbb{R}$ for which $\delta(A, B) = 0$.

(c) Prove that if A is compact, B is closed, and $A \cap B = \emptyset$, then $\delta(A, B) > 0$.

SEE NOTE 88

4.7.12 Show that every closed set can be expressed as the intersection of a sequence of open sets.

4.7.13 Show that every open set can be expressed as the union of a sequence of closed sets.

4.7.14 A collection of sets $\{S_\alpha : \alpha \in A\}$ is said to have the *finite intersection property* if every finite subfamily has a nonempty intersection.

(a) Show that if $\{S_\alpha : \alpha \in A\}$ is a family of compact sets that has the finite intersection property, then

$$\bigcap_{\alpha \in A} S_\alpha \neq \emptyset.$$

(b) Give an example of a collection of closed sets $\{S_\alpha : \alpha \in A\}$ that has the finite intersection property and yet

$$\bigcap_{\alpha \in A} S_\alpha = \emptyset.$$

4.7.15 A set $S \subset \mathbb{R}$ is said to be *disconnected* if there exist two disjoint open sets U and V each containing a point of S so that $S \subset U \cup V$. A set that is not disconnected is said to be *connected*.

(a) Give an example of a disconnected set.

(b) Show that every compact interval $[a, b]$ is connected.

(c) Show that $\mathbb{R}$ is connected.

(d) Show that every nonempty connected set is an interval.

4.7.16 Show that the only subsets of $\mathbb{R}$ that are both open and closed are $\emptyset$ and $\mathbb{R}$.

4.7.17 Given any uncountable set of real numbers E show that it is possible to extract a sequence $\{a_k\}$ of distinct terms of E so that the series $\sum_{k=1}^{\infty} a_k/k$ diverges.

Notes

[73]Exercise 4.2.25. Let $\{q_n\}$ be an enumeration of the rationals. If x is isolated, then there is an open interval I_x containing x and containing no other point of the set. Pick the least integer n so that $q_n \in I_x$. This associates integers with the isolated points in a set.

[74]Exercise 4.3.1. Consider the set $\{1/n : n \in \mathbb{N}\}$.

[75]Exercise 4.3.23. The ternary expansion of a number $x \in [0,1]$ is given as

$$x = 0.a_1a_2a_3a_4\cdots = \sum_{i=1}^{\infty} a_i/3^i$$

where the $a_i \in \{0,1,2\}$. (Thus this is merely the "base 3" version of a decimal expansion.) Observe that $1/3$ and $2/3$ can be expressed as $0.0222222\ldots$ and $0.200000\ldots$ in ternary. Observe that each number in the interval $(1/3, 2/3)$, that is the first stage component of G, must be written as $0.1a_2a_3a_4\ldots$ in ternary. How might this lead to a description of the points in G?

[76]Exercise 4.4.6. Consider the intersection of the family of *all* closed sets that contain the set E.

[77]Exercise 4.4.7. Consider the union of the family of *all* open sets that are contained in the set E.

[78]Exercise 4.5.1. Try this one: Define $f(x) = 0$ for x irrational and $f(x) = q$ if $x = p/q$ where p/q is a rational with p, q integers and with no common factors.

[79]Exercise 4.5.5. Take compact to mean closed and bounded. Show that a finite union or arbitrary intersection of compact sets is again compact. Check that an arbitrary union of compact sets need not be compact. Show that any closed subset of a compact set is compact. Show that any finite set is compact.

[80]Exercise 4.5.8. For a course in functions of one variable open covers can consist of intervals. In more general settings there may be nothing that corresponds to an "interval;" thus the more general covering by open sets is needed. Your task is just to look through the proof and spot where an "open interval" needs to be changed to an "open set."

[81]Exercise 4.5.9. Cousin's lemma offers the easiest proof, although any other compactness argument would work. Take the family of all intervals $[c,d]$ for which $f(c) < f(d)$ and check that the hypotheses of that lemma hold on any interval $[x,y]$.

[82]Exercise 4.5.18. Let $\mathcal{C} = \{V_\alpha : \alpha \in A\}$ be the open cover. Let N_1, N_2, ...be a listing of all open intervals with rational endpoints. For each $x \in E$ there is a V_α and a k so that the interval N_k satisfies $x \in N_k \subset V_\alpha$. Call this choice $k(x)$. Thus

$$\mathcal{N} = \{N_{k(x)} : x \in E\}$$

is a countable open cover of E (but not the countable open cover that we want). But corresponding to each member of $\mathcal{N}$ is a member of $\mathcal{C}$ that contains it. Using that correspondence we construct the countable subcollection of $\mathcal{C}$ that forms a cover of E.

[83]Exercise 4.5.19. Lindelöff's theorem asserts that an open cover of any set of reals can be reduced to a countable subcover. The Heine-Borel theorem asserts that an open cover of any compact set of reals can be reduced to a finite subcover.

[84]Exercise 4.5.20. For (b)⇒(d) and for (c)⇒(d). Suppose that there is an open cover of A but no finite subcover. Step 1: You may assume that the open cover is just a sequence of open sets. (This is because of Exercise 4.5.18.) Step 2: You may assume that the open cover is an increasing sequence of open sets

$G_1 \subset G_2 \subset G_3 \subset \ldots$ (just take the union of the first terms in the sequence you were given). Step 3: Now choose points x_i to be in $G_i \cap A$ but not in any previous G_j for $j < i$. Step 4: Now apply (b) [or (c)] to get a point $z \in A$ that is an accumulation point of the points x_i. This would have to be a point in some set G_N (since these cover A) but for $n > N$ none of the points x_n can belong to G_N.

[85]Exercise 4.6.5. This result may seem surprising at first since the Cantor set, at first sight, seems to contain only the endpoints of the open intervals that are removed at each stage, and that set of endpoints would be countable. (That view is mistaken; there are many more points.) Show that a point x in $[0,1]$ belongs to the Cantor set if and only if it can be written as a ternary expansion $x = 0.c_1c_2c_3\ldots$ (base 3) in such a way that only 0's and 2's occur. This is now a simple characterization of the Cantor set (in terms of string of 0's and 2's) and you should be able to come up with some argument as to why it is now uncountable.

[86]Exercise 4.6.9. You will need the Bolzano-Weierstrass theorem (Theorem 4.21). But this uncountable set E might be unbounded. How could we prove that an uncountable set would have to contain an infinite bounded subset? Consider

$$E = \bigcup_{n=1}^{\infty} E \cap [-n,n].$$

[87]Exercise 4.6.10. Select a rational number from each member of the family and use that to place them in an order.

[88]Exercise 4.7.11. For part (b) look ahead to part (c): Any such example must have A and B unbounded. For part (c) assume $\delta(A,B) = 0$. Then there must be points $x_n \in A$ and $y_n \in B$ with $|x_n - y_n| < 1/n$. As A is compact there is a convergent subsequence x_{n_k} converging to a point z in A. What is happening to y_{n_k}? (Be sure to use here the fact that B is closed.)

Chapter 5

CONTINUOUS FUNCTIONS

5.1 Introduction to Limits

The definition of the limit of a function

$$\lim_{x \to x_0} f(x)$$

is given in calculus courses, but in many classes it is not explored to any great depth. Computation of limits is interesting and offers its challenges, but for a course in real analysis we must master the definition itself and derive its consequences.

Our viewpoint is larger than that in most calculus treatments. There it is common to insist, in order for a limit to be defined, that the function f must be defined at least in some interval $(x_0 - \delta, x_0 + \delta)$ that contains the point x_0 (with the possible exception of x_0 itself). Here we must allow a function f that is defined only on some set E and study limits for points x_0 that are not too remote from E. We do not insist that x_0 be in the domain of f but we do require that it be "close." This requirement is expressed using our language from Chapter 4. We must have x_0 a point of accumulation of E.

Except for this detail about the domain of the function the definition we use is the usual ε-δ definition from calculus. Readers familiar with the sequence limit definitions of Chapter 2 will have no trouble handling this definition. It is nearly the same in general form as the ε-N definition for sequences, and many of the proofs use similar ideas.

5.1.1 Limits (ε-δ Definition)

The definition of a sequence limit, $\lim_{n\to\infty} s_n$, made precise the statement that s_n is arbitrarily close to L if n is sufficiently large. The definition of a function limit

$$\lim_{x \to x_0} f(x)$$

is intended, in much the same way, to make precise the statement that $f(x)$ is arbitrarily close to L if x is sufficiently close to x_0. One feature of the definition must be to exclude the value at the point x_0 from consideration; it should be irrelevant to the value of the limit. It is possible (likely even) that $f(x_0) = L$, but whether this is true or false should not be any influence on the existence of the limit.

Thus the definition assumes the following form. The requirement that x_0 be a point of accumulation of E may seem strange at first sight, but we will see that it is needed

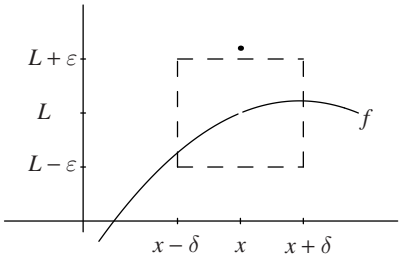


Figure 5.1. Graphical interpretation of the ε-δ limit definition.

in order for the definition to have some meaning. Without it any number would be the limit and the theory of limits would be useless.

Definition 5.1: (Limit) Let $f : E \to \mathbb{R}$ be a function with domain E and suppose that x_0 is a point of accumulation of E. Then we write

$$\lim_{x \to x_0} f(x) = L$$

if for every $\varepsilon > 0$ there is a $\delta > 0$ so that

$$|f(x) - L| < \varepsilon$$

whenever x is a point of E differing from x_0 and satisfying $|x - x_0| < \delta$.

Note. The condition on x can be written as

$$0 < |x - x_0| < \delta$$

or as

$$x \in (x_0 - \delta, x_0 + \delta), \quad x \neq x_0$$

or, yet again, as

$$x_0 - \delta < x < x_0 + \delta, \quad x \neq x_0.$$

The exclusion of $x = x_0$ should be seen as an advantage here. An inequality is required to be true for all x satisfying some condition, and we are allowed *not* to have to check $x = x_0$. It may happen to be true that $|f(x) - L| < \varepsilon$ when $x = x_0$ but it is irrelevant to the definition. For example, you will recall that the limit used to define a derivative

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

must require that the value for $x = x_0$ be excluded; the expression is not defined when $x = x_0$.

See Figure 5.1 for a graphical interpretation of the definition. In the picture a particular value of ε is illustrated and for that value the figure shows a choice of δ that works. Every smaller value of δ would have worked, too. The definition requires doing this, however, for *every* positive ε, and the figure cannot convey that.

We now present some examples illustrating how to prove the existence of a limit directly from the definition. These are to be considered as exercises in understanding the definition. We would rarely use the definition to compute a limit, and we hope seldom to use the definition to verify one; we will use the definition to develop a theory that will verify limits for us.

Example 5.2: Any function $f(x) = ax + b$ will have the easily predicted limit

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} (ax + b) = ax_0 + b.$$

If you sketch a picture similar to that of Figure 5.1 you see easily that the choice of δ is monitored by the slope of the line $y = ax + b$. The steeper the slope, the smaller the δ has to be taken in comparison with ε.

Let us do this for the linear function $f(x) = 10x - 11$. We expect that

$$\lim_{x \to 5} (10x - 11) = 10(5) - 11 = 39.$$

Let us prove this. We need a condition ensuring that the expression

$$|(10x - 11) - 39|$$

is smaller than ε. Some arithmetic converts this to

$$|(10x - 11) - 39| = |10x - 50| = |10|\,|x - 5|.$$

Now it is clear that, if we insist that $|x - 5| < \varepsilon/10$, we will have

$$|(10x - 11) - 39| < \varepsilon.$$

That completes the proof. Better, though, would be to write it in a more straightforward manner that obscures how we did it but gets to the point of the proof more simply:

> Let $\varepsilon > 0$. Let $\delta = \varepsilon/10$. Then for all x with $|x - 5| < \delta$ we have
>
> $$|(10x - 11) - 39| = |10|\,|x - 5| < 10\delta = \varepsilon.$$
>
> By definition, $\lim_{x \to 5}(10x - 11) = 39$ as required.

An alert reader of our short proof will know that the choice of δ as $\varepsilon/10$ took some time to compute and is not just an inspired second sentence of the proof. ◀

Example 5.3: Let us use the definition to verify the existence of

$$\lim_{x \to x_0} x^2.$$

Again the definition gives no hints as how to compute the limit; it can be used only to verify the correctness of a limit statement. To keep it simple let us show that $\lim_{x \to 3} x^2 = 9$. We need a condition ensuring that the expression

$$\left|x^2 - 9\right|$$

is smaller than ε. Some arithmetic converts this to

$$\left|x^2 - 9\right| = |x - 3|\,|x + 3|.$$

If we insist that

$$|x-3| < \varepsilon/M,$$

where M is bigger than any value of $|x+3|$, then we will have $\left|x^2-9\right| < \varepsilon$ exactly as we need. But just how big might $|x+3|$ be? If we remember that we are interested only in values of x close to 3 (not huge values of x), then this is not too big. For example, if x stays inside $(2,4)$, then $|x+3| < 7$. These are enough computations to allow us to write up a proof.

> Let $\varepsilon > 0$. Let $\delta = \varepsilon/7$ or $\delta = 1$, whichever is smaller (i.e., $\delta = \min\{\varepsilon/7, 1\}$). Then if $|x-3| < \delta$ it follows that
>
> $$|x+3| = |x-3+6| \leq |x-3|+6 < 7$$
>
> and hence that
>
> $$\left|x^2-9\right| = |x-3|\,|x+3| < 7\,|x-3| < 7(\varepsilon/7) = \varepsilon.$$
>
> By definition, $\lim_{x\to 3} x^2 = 9$ as required.

The finished proof is shorter and lacks all the motivating steps that we just went through. ◀

In spite of these examples and the necessity in elementary courses such as this to work through similar examples, the main goal of our definition is to build up a theory of limits that can then be used to justify other methods of computation and lead to new discoveries. On occasions we must, however, return to the definition to handle an unusual case.

Exercises

5.1.1 Prove the existence of the limit $\lim_{x\to x_0}(4-12x)$.
SEE NOTE 89

5.1.2 Prove the validity of the limit $\lim_{x\to x_0}(ax+b) = ax_0+b$.
SEE NOTE 90

5.1.3 Prove the existence of the limit $\lim_{x\to -4} x^2$.
SEE NOTE 91

5.1.4 Prove the validity of the limit $\lim_{x\to x_0} x^2 = x_0^2$.
SEE NOTE 92

5.1.5 Suppose in the definition of the limit that the phrase "x_0 be a point of accumulation of the domain of f" is deleted. Show that then the limit statement $\lim_{x\to -2}\sqrt{x} = L$ would be true for every number L.

5.1.6 Recall that in the definition of $\lim_{x\to x_0} f(x)$ there is a requirement that x_0 be a point of accumulation of the domain of f. Which values of x_0 would be excluded from consideration in the limit

$$\lim_{x\to x_0}\sqrt{x^2-2}?$$

5.1.7 Which values of x_0 would be excluded from consideration in the limit

$$\lim_{x\to x_0} \arcsin|x+2|?$$

5.1.8 Prove the validity of the limit $\lim_{x\to x_0}\sqrt{x} = \sqrt{x_0}$.

SEE NOTE 93

5.1.9 Prove that the limit $\lim_{x\to 0}\frac{1}{x}$ fails to exist.

5.1.10 Prove that the limit $\lim_{x\to 0}\sin(1/x)$ fails to exist.

5.1.11 Using the definition, show that if $\lim_{x\to x_0} f(x) = L$, then

$$\lim_{x\to x_0}|f(x)| = |L|.$$

5.1.12 Suppose that x_0 is a point of accumulation of both A and B and that $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$. We insist that f and g must agree in the sense that $f(x) = g(x)$ if x is in both A and B.

(a) What conditions on A and B ensure that if the limit $\lim_{x\to x_0} f(x)$ exists so too must the limit $\lim_{x\to x_0} g(x)$?

(b) What conditions on A and B ensure that if

$$\lim_{x\to x_0} f(x) \text{ and } \lim_{x\to x_0} g(x)$$

both exist then they must be equal.

SEE NOTE 94

5.1.2 Limits (Sequential Definition)

The theory of function limits can be reduced to the theory of sequence limits. This is a popular device in mathematics. Some new theory turns out to be contained in an old theory. This allows easy proofs of results since the old theory has all the tools needed for constructing proofs in the new subject. If our goal were merely to prove all the properties of limits, this would allow us to skip over ε-δ proofs. But since we are trying in this elementary course to learn many methods of analysis, we shall not escape from learning to use ε-δ arguments. Even so, this is an interesting tool for us to use. We can call upon our sequence experience to discover new facts about function limits.

Definition 5.4: (Limit) Let $f : E \to \mathbb{R}$ be a function with domain E and suppose that x_0 is a point of accumulation of E. Then we write

$$\lim_{x\to x_0} f(x) = L$$

if for every sequence $\{e_n\}$ of points of E with $e_n \neq x_0$ and $e_n \to x_0$ as $n \to \infty$,

$$\lim_{n\to\infty} f(e_n) = L.$$

Note. If x_0 is not a point of accumulation of E, then there would be no sequence $\{e_n\}$ of points of E with $e_n \neq x_0$ for all n and $e_n \to x_0$ as $n \to \infty$. Thus once again this is an essential ingredient of our limit definition.

Before we can use this definition we need to establish that it is equivalent to the ε-δ definition. We prove that now.

Proof. **(Definitions 5.1 and 5.4 are equivalent)** Suppose first that

$$\lim_{x\to x_0} f(x) = L$$

according to Definition 5.1 and that $\{e_n\}$, $e_n \neq x_0$, is a sequence of points in the domain of f converging to x_0. Let $\varepsilon > 0$. There must be a positive number δ so that

$$|f(x) - L| < \varepsilon$$

if $0 < |x - x_0| < \delta$. But $e_n \to x_0$ and $e_n \neq x_0$ so there is number N such that $0 < |e_n - x_0| < \delta$ for all $n \geq N$. Putting these together, we find that

$$|f(e_n) - L| < \varepsilon$$

if $n \geq N$. This proves that $\{f(e_n)\}$ converges to L. This verifies that Definition 5.1 implies Definition 5.4.

Conversely, suppose that L is not the limit of $f(x)$ as $x \to x_0$ according to Definition 5.1. We must find a sequence of points $\{e_n\}$ in the domain of f and converging to x_0 such that $f(e_n)$ does not converge to L. Because L is not the limit, there must be some $\varepsilon_0 > 0$ so that for any $\delta > 0$ there will be points x in the domain of f with $0 < |x - x_0| < \delta$ and yet the inequality

$$|f(x) - L| < \varepsilon_0$$

fails. Applying this to $\delta = 1, 1/2, 1/3, 1/4 \ldots$ we obtain a sequence of points x_n with x_n in the domain of f and

$$0 < |x_n - x_0| < 1/n$$

and yet

$$|f(x_n) - L| \geq \varepsilon_0.$$

This is precisely the sequence we wanted since $\{f(x_n)\}$ cannot converge to L. Thus we have shown that Definition 5.4 implies Definition 5.1. ■

Since the two definitions are equivalent, we can use either a sequential argument or an ε-δ argument in our discussions of limits.

Example 5.5: Suppose we wish to prove that $\lim_{x\to x_0} f(x) = L$ implies that

$$\lim_{x\to x_0} \sqrt{f(x)} = \sqrt{L}.$$

We could convert this into an ε-δ statement, which will involve us in some unpleasant inequality work. Or we can see that, alternatively, we need to prove that if we know $f(x_n) \to L$, then we can conclude $\sqrt{f(x_n)} \to \sqrt{L}$. But we did study just such problems in our investigation of sequence limits (Exercise 2.4.16). ◄

Exercises

5.1.13 Prove the existence of the limit $\lim_{x\to x_0}(4 - 12x)$ by converting to a statement about sequences.